

# Systematising Startup Creation in Cambridge

## A Five-Layer Production Model for Retained Innovation

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### **Abstract**

Cambridge produces world-leading science yet struggles to convert this excellence into retained economic power. This paper — the second in the Global Cambridge Institute series — diagnoses the structural, financial, network, and cultural constraints on venture creation in Cambridge, UK, and proposes the Cambridge Startup Production Model: a coordinated, five-layer system (Training, Capital, Production, Currency, and Culture) that embeds entrepreneurship inside the city’s academic and institutional fabric. The model integrates academic incentives, community-based financial instruments, shared venture-foundry infrastructure, a token-based currency layer, and a cultural control layer that aligns status, ritual, and prestige with venture creation. Together, these interventions aim to reduce fragmentation, accelerate commercialisation, and retain the economic value of Cambridge’s scientific output without sacrificing the organic, decentralised character of its innovation ecosystem.

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# 1 PREFACE

The original version of this document was first published on the 5th of December, 2025. Since then, we have had further, valuable discussion with the stakeholder community of Cambridge, UK, and more time since the planned release date of our initial strategic document to release more of the unmentioned concepts we had previously in development. This second version of our report on Systematising Startup Creation is updated with further insight, structure, additional concepts, and some errata fixed to improve the reader experience and deliver a fuller picture of the vision for the future of Cambridge, and beyond.

Adjustments include:

- The addition of graphics throughout the report to improve conceptual clarity
- Framing of the Startup Production Model as filling 4 pillars – integration, rejuvenation, community, and autonomy
- Defining the Startup Production Model as leveraging the introduction of integrated academic structures, capital incentives, cultural reinforcement, and global positioning into a single system with talent allocation
- Introduced a fifth layer to the Startup Production Model, that of Cultural interventions
- Restructured the Startup Production Model according to a 5-layer metamodel, where Training, Capital, and Production are moderated by the Currency layer, which together are moderated by the Cultural layer.
- Elaborated upon the PhD by Innovation

## 2 INTRODUCTION - THE GENERAL PROBLEM

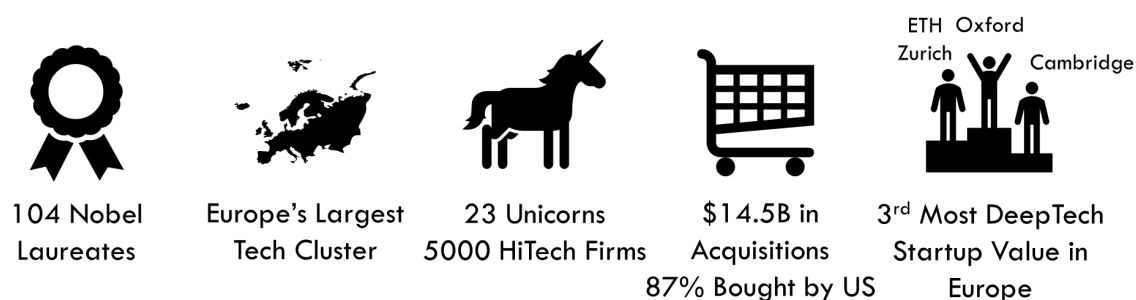


Figure 1: Cambridge's Innovation Paradox

### 2.1 Cambridge's Innovation Paradox

Cambridge produces world-leading science yet struggles to convert this excellence into retained economic power. The city is home to a top university with 104 Nobel laureates (second only to

Harvard) and has given rise to Europe's largest tech cluster, comprising over 5,000 high-tech firms and 23 unicorns (26). However, much of the value created is not staying local.

From October 2023 to 2024, Cambridge companies experienced \$14.5 billion in acquisitions, with 87% purchased by US investors, including major exits such as Abcam (acquired by Danaher for \$5.7 billion) (22) and Darktrace (acquired by Thoma Bravo for \$5.3 billion) (66). This innovation exodus means that although Cambridge-originated ventures succeed, their economic gains frequently flow out of the region. The University of Cambridge is, despite this, 3rd in Europe for creating DeepTech Startup Value, behind the University of Oxford and ETH Zurich (24).

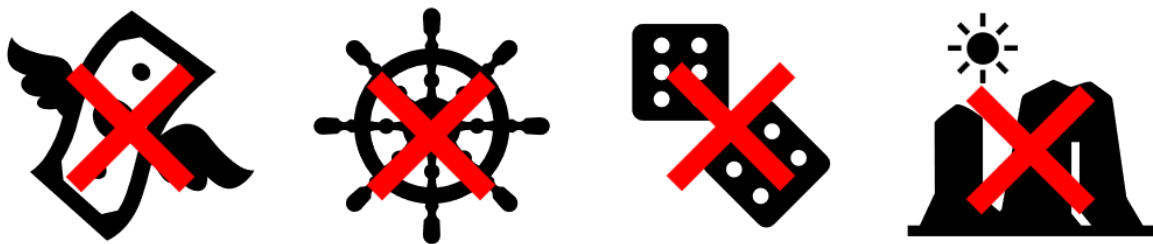
### 2.1.1 Key Structural Constraints

- Cambridge produces world-leading science but fails to convert it into retained economic power.
- The ecosystem functions from the bottom up: there is no central owner, and initiatives tend to appear and disappear as the population cycles.
- Cambridge is often perceived as a place of students and science, rather than a hub for business creation.
- Network topology weaknesses include strong ties internally (within colleges and disciplines) but weak ties across sectors, limiting cross-pollination.
- The result is fragmentation, low retention, and discontinuity in commercial activity.

### 2.1.2 The Innovation Paradox

Where the previous report examined the governance and fiscal instruments needed to stimulate investment in Cambridge, this paper focuses on the production mechanisms of innovation themselves. It proposes a formalised model for automating startup creation, with the aim of converting Cambridge's scientific excellence into retained economic strength.

The approach centres on building an integrated culture, community, and governance mechanism designed to amplify and leverage the existing ecosystem.



**Figure 2:** The Need for Systematisation

## 2.2 The Need for Systematisation

- Serendipity dominates (formals, pubs, random events) → unscalable, unreliable.
- No “*production model*” for turning ideas into companies.
- No shared infrastructure, no data layer, no coordinated portfolio construction.
- Opportunity cost of foundership is enormous (£400k–£600k).
- Cambridge lacks a cohesive narrative of itself as a startup-producing engine.

## 2.3 Clarification

This paper proposes a collection of system-level cultural, financial, and operational interventions to produce a self-maintaining innovation culture. Following the case of Cambridge, UK, the goal is to automate and formalise the creation of companies within an academic innovation ecosystem. The proposed model integrates academic incentives, capital structures, talent allocation, cultural reinforcement, and global positioning into a single coordinated system. In essence, it aims to make startup formation a deliberate process – supported by institutions and data – rather than a fortunate accident. Before detailing the solution, we examine specific problem areas in depth, with supporting evidence, to ground the recommendations in reality

We draw from principles derived from (1), which states that:

Innovation ecosystem interventions... are more likely to succeed when they account for the current state of a given regional ecosystem (latent capacities, current bottlenecks, and economic and institutional constraints) and when they involve extended commitments by multiple stakeholders within that ecosystem.

Accelerating Innovation Ecosystems: The Promise and Challenges of Regional  
Innovation Engines, 2023, National Bureau of Economic Research

Consequently, this report looks to open discussion on these key points, with proposals on how to address them, with mind for follow-ups with in-depth analyses.

The goal is to define a set of interventions to apply to Cambridge and illustrate them as opportunities for enterprising individuals to seize, helping form a unified innovation community. In placing them together, we contextualise them against each other, allowing those who pursue any given intervention to scale their impact to both community creation and their own initiative, increasing the likelihood and outcomes of their growth. In this, we help stimulate, guide, and systematise the development of an autonomous innovation community.

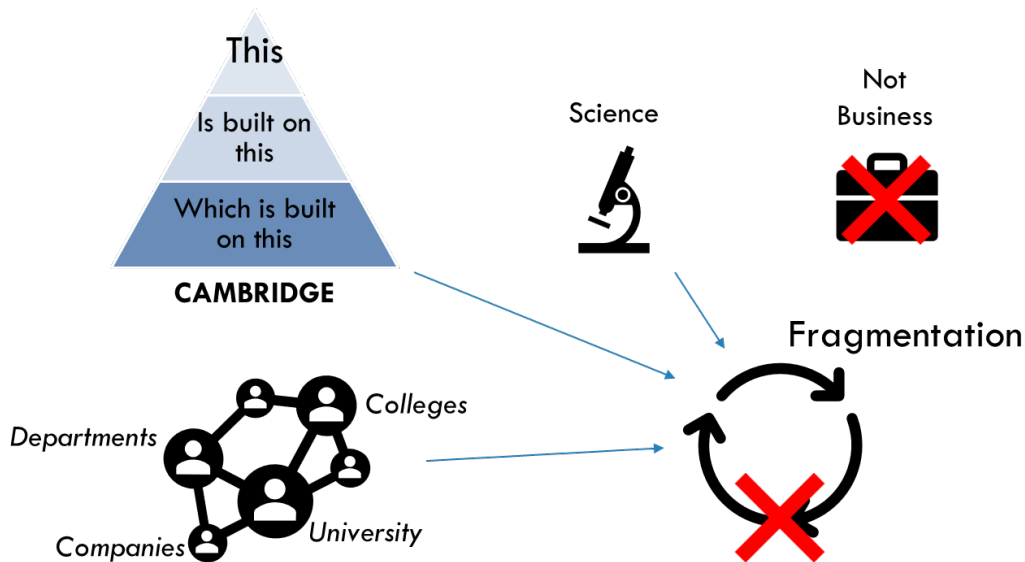


Figure 3: 2.1 Structural Characteristics

## 3 DETAILED PROBLEMS

### 3.1 Structural Characteristics

Cambridge is a system for exporting science to the world – efficient, but unscalable. This is because:

- It is a bottom-up ecosystem, made of independent institutions with no central owner – colleges, university, companies are independent & disorganised, not communicating over silos or uniting initiatives.
- Cambridge’s narrative is a place of students & science, not business, unlike Silicon Valley.
- Cambridge cycles people – students, workers, academics – each spend a short time in Cambridge before leaving.

The result: fragmented growth, low retention; initiatives appear/disappear yearly, & ironically limited learning on how to grow the ecosystem.

Cambridge is fundamentally a bottom-up ecosystem. There is no central coordinating authority, so initiatives emerge only when a sufficient concentration of social and financial capital aligns behind them. Analyses of the “*Cambridge Phenomenon*” describe its emergence as the product of entrepreneurial academics, early technology founders, and college-based networks, rather than any central authority or coordinated regional strategy (18). Today, within a 20-mile radius of the city, more than 150 innovation sites host over 4,000 companies and around 100,000 employees, with the wider Silicon Fen region home to over 5,000 knowledge-intensive firms and over 100,000 cluster jobs (6).

This growth has unfolded atop a highly mobile population. Public-health assessments note that most students are transient, cycling between home and university multiple times per year (17).

Externally, Cambridge is firmly established as a science and startup hub. International reporting has long described the Silicon Fen as Europe’s closest analogue to Silicon Valley, and recent regional positioning frames it as one of the most dynamic innovation centres in Europe. The University of Cambridge estimates its annual UK economic impact in 2023 at £29.8 billion, of which £23 billion (78 percent) derives from spin-outs, closely associated firms, and research-led commercial activity (47). Yet the city’s brand remains anchored in colleges and laboratories, although these narratives are now complemented by founder-centric storytelling through efforts such as the Cambridge Innovation Story campaign (11).

Cambridge’s innovation culture is expressed through a rich tapestry of clubs, hackathons, departmental incubators, and college angel funds, but it faces challenges in maintaining continuity and scale. Programs often end when their champions leave. For example, if a particular professor-advisor retires or a funding grant ends, a successful initiative (such as a startup competition or accelerator) may dissolve, with no guarantee of a successor. Moreover, interconnections between initiatives are weak. A startup bootcamp in one department might have no visibility of a similar program in another. There are limited shared maps of who is doing what across the city.

This produces a system that is highly productive yet increasingly leaky. Cambridge, Oxford and London generate roughly 40 percent of all UK university spin-outs, but Cambridge and Oxford also lose more than 100 each to relocation, more than any other UK cities, according to the Centre for Cities 2025 analysis of spin-out retention (46). At the same time, senior business and civic leaders repeatedly highlight housing, transport, and water constraints as key factors limiting the city’s ability to retain scaling firms and sustain long-term growth (52). Thus, although Cambridge leads the UK in scientific output and innovation-driven economic activity, a substantial portion of the value it creates continues to be captured elsewhere, driven by infrastructure pressures, spatial constraints, and the path-dependent structure of its bottom-up ecosystem.

Despite many pieces in place, Cambridge does not have a unified startup production line. There is no single framework that links education, research, funding, and incubation into one continuum for entrepreneurs. Currently, a would-be founder must navigate a fragmented landscape. They might take an entrepreneurship course, separately find a mentor, independently apply to an accelerator, and so on, with each step uncoordinated. The absence of a system owner or integrator means opportunities are missed and efforts duplicated. For example, five different accelerators may each recruit five similar AI startups rather than pooling resources to ensure the best 25 AI ideas in Cambridge all receive support in a tiered structure for ideation, seed, and growth stages. Top ecosystems often have clearer pathways. Stanford’s entrepreneurship tracks or MIT’s layered programs guide founders from dorm-room idea to venture funding within an organised support sequence (35). Cambridge can adopt some of that coherence to accelerate its innovation rate.

A related structural issue is Cambridge’s network topology, which exhibits strong bonding ties

but weak bridging ties. Within closed groups, such as researchers in the same field or alumni of a college, we see high homophily in the ecosystem, where relationships are tight and trust is high. This supports excellent knowledge exchange within silos, but it limits diversity of input. Meanwhile, connections between academia and industry, or across disciplines, have low heterophily, being sparse and transactional. A 2023 Cambridge Ahead study highlighted the importance of clustering business parks next to academic sites to spur interaction (5), yet cultural and institutional barriers often limit this. As noted in the previous report, the Cambridge ecosystem is small in population but globally connected. Paradoxically, this makes it easy for local talent to collaborate with outsiders (for example, by shifting to opportunities in London or overseas) rather than breaking silos at home. Any solution must therefore account for Cambridge’s decentralisation and design opt-in, non-disruptive interventions that encourage collaboration without distracting people from their core, often independent, pursuits.

In summary, Cambridge’s structural DNA – bottom-up growth, autonomous sub-communities, and transient talent – has created a fragmented and intensely active environment. There is tremendous innovation happening, but it is not being aggregated into a cohesive engine. Without continuity and stronger cross-cutting networks, the city cannot fully leverage its breadth of expertise. Interventions must therefore operate at the system level, introducing connective tissue and institutional memory into this highly federated ecosystem.

### 3.2 Innovation Consolidation Problem

- Science moves fast; commercialisation moves slow
- Incentives are misaligned because researchers are rewarded more for publishing than building

Innovation in Cambridge progresses so rapidly that consolidation has been largely neglected. People are focused, and expected to focus, on scientific output. Researchers are incentivised to publish or patent rather than scale their ideas. One major cause is that academic career pathways reward publication rather than commercialisation.

A commonly cited concern is a culture of creating spin-outs for the sake of spinning out. The British Venture Capital Association (BVCA) has argued that universities sometimes form companies too early in an idea’s life (36). Jo Taylor, chair of the BVCA’s Tech Committee, notes that *“some of these university companies are spun out as projects, rather than proper businesses.”*

There is also a shortage of dedicated translational roles – people whose job is to proactively turn science into startups. Cambridge has relatively few entrepreneurs in residence embedded in labs or departments (10), with only 2 compared to, for example, Lancaster with over 90 (44). MBA students and professional managers sometimes attempt to fill this gap, but it remains unclear whether an MBA is the optimal interface for deep science ventures given academic workloads. In the current setup, faculty and PhDs are expected to suddenly become startup

CEOs, a leap for which they have far less training than for their scientific work.

The absence of formal innovation concierge roles means that promising ideas often languish until a chance encounter connects a scientist with a business counterpart, unless the scientist already has the business awareness to engineer such a connection. This falls under the challenges of serendipity, which, although well engineered in Cambridge (12), still create an inefficient pipeline. Cambridge’s science moves fast, but translating those advances into market-ready enterprises moves slow.

Another dimension is the industry versus academia talent flow. Cambridge has world-class researchers who may not be natural CEOs, and it has access to industry veterans (including those at Science Park companies or alumni in London) who could be exceptional executives but are not sufficiently engaged. This creates a misallocation where some researchers reluctantly become underprepared CEOs, while experienced managers remain in corporate roles instead of leading high-impact startups.

A refined system would allow leadership talent to be injected into research-born startups at early stages without alienating founders. Some U.S. accelerators, such as Y Combinator, explicitly teach technical founders how to become businesspeople quickly. Cambridge could adopt similar approaches or establish co-founder matchmaking mechanisms. Currently, there is no formal clearinghouse for people seeking co-founders across the University and wider community, which limits effective talent alignment.

Cambridge has strong research groups and data analysis units, but it lacks a dedicated knowledge infrastructure for the startup ecosystem. There is no platform or institution that systematically gathers and disseminates operational knowledge about startups (not just knowledge created by startups). For instance, there is no well-known central repository of Cambridge-specific venture playbooks, case studies, or lessons learned. Silicon Valley benefits from an ongoing feedback loop of blogs, events, and alumni mentorship that continually transfers knowledge to new founders. Cambridge could benefit from a more systematic mechanism such as a Cambridge Startup Handbook updated annually. Although judges and coaches exist, knowledge sharing is mostly informal or limited to one-off events. A dedicated knowledge hub – physical or digital – that curates best practices (for example, navigating university IP, accessing lab space, identifying the right contacts) would significantly reduce friction for founders.

### **3.2.1 Unicorn Bias, Exodus Bias, and Low Retention**

There is a cultural tendency to focus on big wins, which can undermine steady innovation consolidation. Cambridge, like many elite research centres, celebrates the occasional unicorn or lucrative IP licensing deal. However, this emphasis can encourage founders to leave Cambridge to scale, for example by moving headquarters to London or San Francisco where late-stage capital is more abundant.

The focus on unicorns or quick exits creates a support gap for moderate-growth companies, which might otherwise become stable mid-sized employers anchoring the region. These ventures receive less attention amid the unicorn narrative. When ambitious spin-outs feel obliged to grow big or leave, many ultimately leave – either by selling early or relocating for the next growth stage. This exodus bias contributes to low retention of successful ventures.

A recent analysis noted that Cambridge is rapidly challenging London as a tech hub, yet must mature in late-stage capital and scale readiness if it is to avoid losing companies at the breakout stage (67). Without this, companies that could become anchors instead exit the local ecosystem. Low retention has significant second-order effects. It diminishes:

- the pool of experienced entrepreneurs
- the locally re-investable wealth
- the density of mentors and repeat founders

Silicon Valley thrives on a virtuous cycle where successful founders reinvest as angels or mentors. Cambridge could capture more of this if departures were less frequent. Alumni donations benefit colleges and research, but they do not fund new startups or reward innovators directly, and they largely exclude non-university innovation. Venture groups such as the Cambridge Angels help, but they are not optimised for mid-cap ventures or those at the intersection of academia and commercial innovation.

One positive example is that some large R&D companies entrenched in Cambridge reinvest through partnerships and local hiring, such as AstraZeneca with its £1 billion global research HQ (26). However, this is a narrow channel of impact compared to the broad recycling of talent and capital found in more self-sustaining ecosystems.

To truly consolidate innovation, Cambridge must retain more of its entrepreneurial output. Companies and talent need to remain in the region long enough to create a self-reinforcing, self-sustaining innovation cycle.

### 3.3 Funding Pipeline Gaps

- Early commercial research is underfunded
- Spin-outs are often projects rather than companies

As a consequence of the tentative approach to entrepreneurship observed among operators in the UK, founding teams often rush to incorporate and seek equity funding before fully investigating market needs or developing a viable product. Combined with Cambridge’s structural limitations and immature formal mechanisms for funding early commercial exploration, this leads to inconsistent deal quality: startups based on brilliant science but lacking product-market fit or committed leadership. Taylor and the BVCA advocate for more proof-of-concept research funded by grants (not equity) before forming a company (36): “*More*

*work, funded by grant rather than equity, should go into researching an idea's commercial potential before forming a company around it.”*

Cambridge, and the UK more broadly, suffers from funding gaps at the earliest stages of commercial development. There is ample funding for pure research (through grants and government R&D budgets) and significant venture capital for proven startups, as evidenced by Cambridge's £2.3 billion in VC funding in 2024 (67). The difficulty lies in the middle: the translational “*valley of death*,” where an idea is too applied for research grants but too unproven for venture capital, occupying the first chasm of the **triple chasm concept** pioneered within Cambridge (53).

University seed funds and local angels attempt to fill this space, but their resources are limited. Founders frequently incorporate companies simply to chase small pots of money (a £50k grant or a £100k angel cheque) when what they actually need is deeper research or market exploration. Premature company formation can deter later investors; a project masquerading as a startup may falter during due diligence, reinforcing inconsistent deal quality.

Currently, Cambridge offers few formal mechanisms for pre-company funding. However, some initiatives are emerging: Cambridge Enterprise and university partners have begun offering Founders at Cambridge grants and a new Technology Investment Fund for very early-stage exploration (9). An independent review noted that Cambridge increased its spinouts to 26 in 2024, which was more than Oxford or Imperial that year, and the East of England led all UK regions with 35 percent of total spin-out investment (9). However, these figures were buoyed by a small number of mega-deals, such as Bicycle Therapeutics raising £450 million (3), highlighting a persistent pattern: a minority of high-profile spin-outs attract large sums, while many smaller ideas struggle to secure any funding.

Work is underway to introduce a Cambridge Stock Exchange (16), intended to allow equity trading at pre-IPO stages. This could create a dynamic system of transactions in sectors like biotech. However, it applies to only one sector, and to upgrade a community, long-term commitment mechanisms must be implemented. An exchange alone does not reward commitment directly.

Despite these emerging mechanisms, the absolute funding levels remain modest relative to scientific output. Cambridge's tech ecosystem had a strong year in 2024, nearly doubling funding from 2023, yet Dealroom's analysis found that London still raised five times more capital (£8.4B versus Cambridge's £1.7B in 2024) (23). While Cambridge punches above its weight in efficiency—delivering 16.9× value per VC dollar (far above the UK average), the funding gap remains especially visible in the proof-of-concept phase.

Part of the challenge is that early-stage investors encounter inconsistent deal flow. Cambridge innovators' approach to business contributes to this: some ventures are world-class, while others lacking pre-company vetting and appear half-formed (43). This inconsistency can dissuade major investors from dedicating attention to Cambridge, creating a self-reinforcing cycle of

limited capital. Cambridge therefore needs a more reliable feeder system, including small-scale commercial research grants, incubators to validate ideas, and knowledgeable seed funds capable of bridging university projects to Series A-ready companies.

This may be described as the absence of a citywide funding grid or fund-of-funds ensuring coverage across stages and sectors. Multiple funds exist, but there is no coordination to guarantee that a promising startup does not fall into a gap between a grant and seed round, or seed and Series A. The idea of a Community Fund – capital contributed by colleges, local authorities, corporates, and alumni, has been suggested but not implemented. Such a fund could co-invest in every credible Cambridge startup, effectively indexing the local innovation economy. Returns could be recycled into new ventures, creating a virtuous funding cycle. Without it, Cambridge relies on external investors who have no inherent loyalty to the region; they will prioritise their own fund performance, which may mean relocating companies or pushing early exits. A community fund could better align incentives to keep companies local.

Additionally, no shared incentive structure exists. Local stakeholders like colleges or councils do not gain directly if a startup succeeds, so engagement is limited. A funding grid with tokenised equity stakes for community institutions could change that. Israel’s Yozma programme partially nationalised the upside of startups and reinvested gains into new funds. Cambridge could emulate a smaller version; for instance, a small percentage of equity from university spin-outs could go into a collective pool. This kind of risk-pooling and incentive alignment is currently missing.

Cambridge’s startup portfolio is fragmented and uncoordinated at the ecosystem level, primarily because ventures are backed by disparate actors such as individual college endowments, angel networks, and VC funds, each optimizing for independent returns rather than citywide outcomes. In the absence of a unified investment strategy, capital allocation becomes suboptimal; multiple groups may competitively chase “*trendy*” sectors like AI or biotech, while core Cambridge strengths in manufacturing science or social innovation remain underfunded. This fragmentation also means the ecosystem lacks shared risk-mitigation instruments—such as a common fund or citywide seed consortium, that could hedge bets across the region and ensure promising ideas do not die simply from a localized lack of capital (62).

Cambridge Innovation Capital (CIC) operates as a quasi-local venture capital fund. However, until recently, CIC was relatively modest in scale. In 2025, CIC announced a £100 million commitment for university spin-outs as part of a £250 million fund (70), signalling an important step toward strengthening Cambridge’s capital infrastructure. Even so, broader community-wide coverage, such as a shared innovation fund or open co-investment scheme accessible to all innovators, remains absent. As a result, Cambridge’s innovation assets resemble pieces of a jigsaw yet to be assembled. Addressing this will require structural solutions that encourage portfolio-level thinking and resource-sharing across the ecosystem, which could significantly improve outcomes. Combined with Cambridge’s bottom-up structure, which focuses primarily on generating innovation rather than simplifying or integrating the processes

around it, the absence of a system-level portfolio approach results in:

- No hedging or co-investment structures across the city
- No cumulative data infrastructure for tracking venture performance
- No instruments for sharing risk at the ecosystem level

### **3.4 Network Topology Weaknesses**

- Strong internal ties, weak cross-sector ties
- Low diversity in trusted networks; high diversity in weak-trust networks

Any solutions must recognise that Cambridge is highly decentralised, and most actors in the city are focused on delivering their own projects to external customers. The local ecosystem is small, but its level of external contact is high, creating many opportunities to pivot outward. Therefore, any intervention must:

- Not distract
- Not constrain
- Be optional or opt-in
- Be sufficiently valuable
- Have short paths to value, allowing participants to quickly return to their core work

#### **3.4.1 Cambridge's Network Topology Problem**

Cambridge exhibits an unusual network topology:

- Within disciplines (colleges, departments): extremely strong ties, high trust, low diversity
- Between academic and commercial nodes: weak ties, high diversity, low trust

This creates collaboration paradoxes. To resolve them, Cambridge needs infrastructure for productive weak ties – connections strong enough for knowledge transfer but weak enough to preserve network diversity.

As mentioned, Cambridge's network is marked by strong internal ties and weak cross-sector ties, which poses a significant challenge to innovation. Within colleges, departments, and research groups, people trust each other and collaborate deeply, but these groups are often insular. For example, a team of engineers might interact daily with other engineers yet never meaningfully engage with the business school or local industry managers. This produces low diversity of input within trusted networks, even though Cambridge overall contains extraordinary intellectual breadth. Conversely, where cross-sector interactions do occur (for instance, a tech founder meeting a biotech professor at an event), they tend to show low trust, because the individuals operate in different circles and lack pre-existing relationships.

This “*two-layer*” network defined by dense clusters and separated by sparse bridges creates a collaboration paradox. While breakthrough innovation often requires interdisciplinary cooperation (such as medics and computer scientists co-developing AI for healthcare), these connections frequently fail to form or persist. Despite the vast potential for partnership in Cambridge, the social infrastructure required to sustain productive “*weak ties*” remains underdeveloped. A productive weak tie is a connection that is not closely partnered (thus offering new information) but still strong enough to act on, with enough trust to collaborate (45). Cambridge needs more venues and platforms that encourage precisely these interactions, where academics, founders, and corporate partners intersect regularly in a semi-structured, repeatable way.

Cambridge undeniably has world-class technical talent, but it is not optimally paired with operational and commercial talent. Many technically brilliant founding teams lack a CEO-type or product manager, while MBAs with startup ambitions struggle to find technical co-founders. The ecosystem does not yet have an effective mechanism for matching complementary talent pools.

In Silicon Valley, it is common for a business-minded person to team up with an engineer they met at a hackathon or through a dense network of community events. In Cambridge, those networks are weaker and less integrated. Additionally, Cambridge’s immigration and hiring environment can make it difficult to bring in experienced scale-up executives, meaning startups often miss the timely influx of seasoned leadership they need. This can lead to avoidable missteps in scaling, product development, or market entry.

In essence, Cambridge has the raw talent ingredients but lacks a robust talent marketplace or exchange that mixes them into effective founding teams.

### 3.5 Physical Infrastructure Misallocation

- Lab space: supply does not equal allocation
- No matching mechanism

The physical layout of Cambridge contributes to the unstructured formation of connections. The University and science parks are spread across the city, and until recently there was no central innovation district resembling Kendall Square in Cambridge, Massachusetts. Kendall Square is a dense neighbourhood deliberately mixing labs, startups, corporates, housing, and social spaces (60). It hosts over 130,000 life sciences jobs, about six times the number in Greater Cambridge UK, and has attracted billions in investment by collocating talent and capital (60). Cambridge UK is now moving in this direction with developments such as the Cambridge Biomedical Campus expansion and new innovation centres, but historically its innovation actors have been geographically and institutionally separated.

Within the City, distances between key innovation nodes reinforce this fragmentation. The Judge Business School is 4 km from the Department of Computer Science and Technology, and

4.7 km from the Biomedical Campus, despite Cambridge hosting 560 plus life science companies and 2,700 plus IT and telecoms companies. These are impressive numbers, yet Cambridge is listed as only the 66th best innovation cluster globally (81), despite being the second most intensive, illustrating a spatial coordination problem.

Addressing these network weaknesses does not require forcing everyone into a single location, as decentralization is intrinsic to Cambridge’s identity. Instead, solutions should focus on creating “*connective tissue*” through structured mentorship networks (pairing scientists with business mentors outside their immediate circles), rotational programs (embedding students in industry and industry professionals in academic labs), and city-wide meetups centred around thematic challenges like AI for science. Although Cambridge already hosts many entrepreneurial events, these currently rely on serendipity rather than formal mechanisms. By building infrastructure for productive weak ties, the city can provide enough structure to enable trust and knowledge transfer without suppressing the creative diversity of its many niches. This balance would preserve the strengths of Cambridge’s tight-knit communities while unlocking the interdisciplinary collaboration essential for modern innovation.

This lack of coordination extends into the misallocation of physical infrastructure, specifically lab spaces. While the lab take-up by the end of September 2025 reached 63,600 square feet, a 33% year-on-year decline, an additional 98,200 square feet under offer indicates that total take-up will likely exceed the five-year annual average by 16% (27). However, these strong figures mask a distribution crisis. Space is currently allocated largely through market forces or personal networks, allowing well-funded scale-ups and Big Pharma to secure prime labs while cash-constrained startups struggle. There is currently no dynamic matching mechanism or comprehensive central exchange to prioritize a promising two-person spin-out over an established firm that could operate elsewhere.

The result is a market characterized by misutilization rather than a simple lack of supply. A corporate entity may hold excess lab capacity in anticipation of future growth while a neighbouring biotech startup is forced into unsuitable makeshift spaces. To resolve this, Cambridge must introduce mechanisms to reduce barriers and increase usage dynamism, such as shared lab facilities, time-sliced access, or subsidized startup labs with lower rents. By moving away from a rigid, market-only allocation model and toward a more fluid system of space sharing, the city can more effectively support the innovators who rely on these physical environments for prototyping and experimentation.

### 3.6 Foundership Opportunity Cost

- Huge financial and career losses inhibit entrepreneurship

The affordability of physical and professional space is inextricably linked to access to capital, creating a prohibitive opportunity cost for those beginning the founder’s journey. Individuals with the skill profiles most predictive of entrepreneurial success (technical depth, commercial

intuition, and analytical capability), typically face highly lucrative alternatives in strategy consulting, quantitative finance, or deep-tech leadership. With starting salaries ranging from £45k to £100k (4; 55) and typical annual increases of 10 to 20 percent, a prospective founder may forgo between £275k and £745k in cumulative earnings and career progression over just a few years. Forgoing this stability to build a startup on a shoestring stipend means sacrificing not only liquid income but also pensions and home-ownership potential during prime early-career years.

This financial calculus is intensified by Cambridge’s high cost of living, where house prices have surged by 78 percent over the last decade while wages have grown by only 23 percent (54). Consequently, many talented individuals rationally choose secure employment over high-risk ventures. At present, Cambridge offers limited offsets to this cost; most college entrepreneurship grants are in the £10k to £20k range, which is largely symbolic when contrasted with the actual salary gap. Furthermore, institutional barriers persist; as reported by *Times Higher Education* (May 2025), commercialization is often still perceived as a threat to academic independence, while outdated performance metrics continue to prize publication counts over real-world innovation outcomes (40).

Beyond finances, the psychological and social opportunity costs remain significant. A PhD student or postdoc who leaves academia for a venture may lose eligibility for prestigious fellowships or face a perceived loss of status within traditional academic circles. Without cultural shifts that validate entrepreneurship as a prestigious career path, and without financial instruments to de-risk the transition, Cambridge will continue to see its brightest minds opt for established corporate tracks.

To unlock its full potential, the Cambridge ecosystem must move toward the models seen in competing hubs, such as ETH Zurich’s Pioneer Fellowships, which provide full-time salaries of up to CHF 180k to cover opportunity costs during the riskiest 12 to 18 months of research commercialization (30). By reducing the personal downside through stipends or academic re-entry pathways, Cambridge can significantly increase its venture creation rates.

### 3.7 Invisible Infrastructure Gaps

Cambridge relies heavily on serendipity as a driver of innovation. Formal college dinners, student society mixers, informal lab interactions, and chance encounters in shared spaces such as pubs or streets all contribute to cross-disciplinary contact. Cambridge lore even celebrates how Francis Crick and James Watson announced the DNA discovery in The Eagle pub in 1953. But relying on luck and informal networks is unscalable and unreliable as a venture-production model. Unlike ecosystems with deliberate pipelines, Cambridge lacks a startup production line. There is no standardised process or shared platform to systematically turn research ideas into companies. Promising ideas can languish or advance haphazardly depending on whether founders happen to meet the right mentors or investors.

Other innovation clusters demonstrate that successful systematisation requires tiered venture progression, shared services infrastructure, coordinated production structures, and a balance between structure and autonomy. Any sustainable solution for Cambridge must align the disparate interests of academics, investors, and founders while addressing the often-overlooked dimension of social and psychological welfare. This is a critical necessity as Cambridge has the second highest income inequality in England and Wales (8). Without robust support systems, the personal stresses of entrepreneurship may cause founders to abandon the startup pathway entirely, resulting in a permanent loss of innovation potential.

These challenges constitute **social infrastructure gaps** and include:

- insufficient emotional support for founders
- lack of peer mentorship matching
- no structured decompression pathways
- limited mental-health resources tailored to founder stress

More work will be required to define the specific needs of Cambridge’s founder population.

Global peer ecosystems with more structured approaches illustrate what Cambridge lacks. For example, MIT and Stanford operate tiered entrepreneurship systems that guide innovations from lab to market. At MIT, students can access the Sandbox Innovation Fund (seed grants up to \$25k with mentorship) (48), then progress to The Engine, an MIT-linked accelerator providing specialised lab space, equipment, and venture funding for “*tough tech*” startups. Stanford supports founders through programs like StartX, an accelerator with a community of 1,300 plus companies, and deep ties to Silicon Valley venture capital (63).

Even in Europe, centralised ecosystems such as Station F in Paris provide a concentrated and structured environment that speeds up venture creation. Station F hosts 1,000 plus startups on one campus with integrated support services and defined pathways from idea to scale (64). These examples highlight that leading innovation hubs do not leave startup formation to chance; they build repeatable infrastructure to propel ventures forward.

These examples underline that Cambridge’s heavy reliance on serendipity leaves significant invisible infrastructure gaps. To remain competitive, Cambridge must build structures that make innovation repeatable, reliable, and scalable, while preserving the strengths of its decentralised ecosystem.

### 3.8 Slow Investment Cycles

One contributing factor to Cambridge’s funding landscape is the collegiate funding logic that has historically shaped local capital behaviour. Older Cambridge colleges, supported by centuries of alumni endowments, operate with long-term horizons and a focus on careful stewardship. They are not accustomed to rapid, high-risk investment cycles; their funding style

has traditionally been conservative, built on alumni donations and legacy wealth. This mindset may subtly influence the wider investment culture in Cambridge.

In contrast, younger colleges and newer funds rely more heavily on demonstrating impact, attracting external partners, and proving their value through visible innovation outcomes. They tend to be more ambitious, more flexible, and more oriented toward adaptive, modern investment strategies. This internal dichotomy (traditional vs. modern funding approaches) may help explain why some ventures in Cambridge experience slow or uneven backing.

We summarise this as follows:

### 1. Older colleges

- Rely primarily on alumni donations
- Benefit from familial and historical connections, legacy wealth, bequests, and long-established prestige

### 2. Younger colleges

- Rely more on investments and demonstrable impact
- Have limited inherited donor bases and therefore cultivate industrial capital, innovation programmes, and external partnerships
- Are more ambitious and structurally flexible

## 3.9 Measurement Problem

- No ecosystem-level KPIs
- No monthly index
- No founder-retention tracking
- No data to construct tiers, funding floors, or growth acceleration

Cambridge lacks effective ecosystem-level KPIs and real-time metrics. According to Peter Drucker, “*You cannot improve what you don’t measure*” (82). Currently, Cambridge has no equivalent of a Dow Jones-style index for its startup ecosystem – no monthly or quarterly benchmark tracking the health of innovation activity (for example, new startups founded, aggregate funding levels, talent inflows, or spin-out velocity).

Various organisations publish occasional reports: Cambridge Ahead releases annual economic data, the University tracks spin-out numbers, and platforms such as PitchBook or Dealroom provide funding snapshots. However, these indicators are fragmented, periodic, and lagging. Existing mapping efforts from the Cambridge Cluster Map to departmental dashboards and college-level entrepreneurship reports, offer some value by illustrating activity levels. Yet they fail to capture retention rates, inter-startup customer flows, collaboration networks, or

ecosystem-wide capital efficiency.

Addressing this gap would require a shared data infrastructure across the Cambridge ecosystem. In its absence, Cambridge cannot view itself as a coherent portfolio. It cannot easily identify where ventures stall or relocate, which sectors are saturated or underserved, or whether startups are purchasing services from and partnering with one another.

Without aggregated, longitudinal data, structural bottlenecks remain largely invisible. The absence of a unified dashboard means policymakers and investors often operate with limited visibility into the true dynamics of the ecosystem.

For instance, it would be valuable to know:

- the founder retention rate year to year (how many serial founders build their next venture in Cambridge)
- how capital efficiency (value created per £ invested) changes over time
- which sectors are experiencing increases in talent, IP, or venture formation

These data points exist in pieces but are never integrated. By contrast, ecosystems such as Singapore and Israel track innovation metrics continuously and publish them as part of national innovation strategy. The lack of agreed-upon metrics also makes it difficult to set goals or evaluate interventions. If Cambridge launches a new accelerator or funding programme, by what metric is success judged? Currently, evaluation is ad hoc. Lack of data transparency can further hide emerging problems – for example, if capital inflow fell one quarter, or if a critical talent group (such as AI PhDs) began leaving at higher rates, this might go unnoticed until far too late.

Although the measurement problem is widely acknowledged, it remains unsolved. Cambridge still has no real-time KPIs or dedicated ecosystem dashboard. No entity has assumed the role of an innovation analytics provider for the city. This means the community cannot yet answer even basic questions such as “*Are we doing better this quarter than last?*” with quantitative confidence. Partial efforts such as cluster maps or annual reports are insufficient; a dedicated, regularly updated index is required. The absence of metrics also creates a lack of accountability: if a policy aims to improve, for example, female entrepreneurship, there is currently no mechanism to track progress except through occasional retrospective studies.

In summary, the Measurement Problem reinforces all others. It is difficult to rally support (a narrative challenge) or optimally allocate resources (funding and infrastructure challenges) without solid data. Any systemic solution should include the creation of a Cambridge Innovation Observatory to continually collect, analyse, and publish key indicators. This would enable informed decision-making and help market Cambridge’s progress with credible, evidence-backed storytelling.

## 4 ASSETS AND EXISTING INITIATIVES

Despite the challenges above, Cambridge possesses formidable assets on which to build a systematic startup engine. These include institutional strengths, programs, and emerging models that can be scaled up or integrated.

### 4.1 Institutional Knowledge and Prestige Assets

- Colleges with centuries of prestige
- Cambridge rituals, formals, social architecture
- Large alumni network
- Academics

The University of Cambridge's college system – with some colleges dating back centuries, provides a unique combination of institutional prestige, deep networks, and social architecture. Colleges such as Trinity, St John's, and King's possess global reputations, substantial endowments, and alumni networks whose names alone open doors. Newer colleges such as Hughes Hall and Lucy Cavendish tend to be more nimble, innovation-oriented, and investment-focused, leveraging their youth and fundraising models to experiment with new approaches. This mixture of old and new is a structural strength.

- Older colleges confer credibility, heritage value, and patient capital.
- Younger colleges pilot novel educational and entrepreneurial models, moving quickly when opportunities arise.

Cambridge can draw on its rituals and social architecture – formals, seminars, and college gatherings, to embed entrepreneurship into traditional settings. For example, a formal dinner could incorporate themed discussion sessions where founders share ideas alongside academic discourse, weaving innovation into long-standing cultural practices. The University is the primary generator of researchers and innovators in Cambridge, and the colleges are the independent but interlinked stakeholders who shape that output. They are therefore critical levers for influencing innovation in the city. Below is an assessment of several younger, innovation-leaning colleges, which tend to be strong candidates for partnership due to their structural flexibility and outward-looking orientation.

#### 4.1.1 Lucy Cavendish College

- Strong recent expansion of innovation-focused programmes
- High proportion of mature and interdisciplinary students
- Proactive stance on widening participation and entrepreneurship
- Administrative flexibility driven by its institutional structure

#### **4.1.2 Homerton College**

- The Changemakers programme breaks down cross-disciplinary barriers
- Strong Education Faculty provides high-quality entrepreneurship teaching capability
- Greater physical space availability compared to central colleges
- Youth and ambition enable experimentation with new models

#### **4.1.3 Clare Hall**

- Visiting Fellow and Innovation Fellow programmes can be expanded
- Existing structures already integrate external experts into the college environment
- Well-suited to hosting entrepreneurial fellows linked to innovation initiatives

#### **4.1.4 Hughes Hall**

- Known for applied, policy-relevant, and professionally oriented programmes
- Operates an established Entrepreneurship and Innovation Hub
- Strong alignment with translational research and real-world impact

#### **4.1.5 Wolfson College**

- One of the largest and most international postgraduate communities in Cambridge
- Active innovation culture and support for interdisciplinary research
- Ample physical space and institutional flexibility for experimental programmes

In general, these attributes suggest that younger, ambitious colleges should be prioritised as strategic partners in any citywide innovation initiative.

### **4.2 Existing Educational Assets**

- MBA secondments
- MET rotations
- Departmental industrial partnerships
- Tutor ecosystem

Cambridge's alumni base is enormous and globally influential – a distributed network of leaders across academia, industry, and government. If mobilised systematically, alumni could become a powerful force as mentors, investors, advisors, and door-openers for startups. Some initiatives exist, such as Cambridge Angels and occasional alumni venture competitions, but there is significant scope to formalise an Alumni Ventures Fund, structured mentorship pools, or

repeatable founder–alumni matching. Leading ecosystems such as MIT and Stanford benefit enormously from alumni who return as investors or supporters; Cambridge, with its global alumni footprint, could replicate this with the right narrative and infrastructure. Even simple steps – such as colleges tracking alumni entrepreneurs and engaging them as mentors – could yield substantial benefits.

The core of Cambridge’s asset base is its researchers and knowledge base. The University consistently ranks among the very top globally for research output and quality. This produces a continuous stream of breakthroughs that could seed new companies. Cambridge’s challenge is not generating IP; it is already a powerhouse of patents and publications. The key challenge is capturing and translating the IP into ventures. Many departments already operate entrepreneurship initiatives, but these efforts remain fragmented rather than coordinated. Cambridge currently has several educational programmes and mechanisms that contribute to venture formation, listed now.

#### **4.2.1 MBA Programmes and Secondments**

- The Judge Business School’s MBA and MFin programmes embed students in local startups through the Cambridge Venture Project, internships, and consultancy engagements.
- These MBA secondments inject short-term commercial expertise into science-led ventures.
- Scaling such programmes could create more hybrid talent – individuals fluent in both technical and managerial domains.

#### **4.2.2 Master of Engineering (MET) and Entrepreneurship Training**

- The MET programme includes business modules and requires students to prototype product ideas, blending engineering and entrepreneurship.
- Other programmes offer electives in technology policy, innovation management, and adjacent areas.
- Student societies such as CUTEC run startup competitions and hackathons that act as informal training grounds for future founders.

#### **4.2.3 Departmental Industrial Partnerships**

- Many departments maintain industry liaison offices or collaborate with corporate partners.
- These partnerships could become structured channels to:
  - pass industry-defined problem statements to students
  - sponsor students pursuing commercially relevant research

- connect founders with early customers or validation partners

#### **4.2.4 The Tutor Ecosystem**

- Cambridge’s pastoral tutor system ensures every student has a faculty advisor.
- With minimal training, tutors could serve as first-line entrepreneurial encouragers, identifying students with startup potential and directing them to relevant resources.
- This system already exists; it requires only light adaptation to become a distributed early-stage founder-support network.

#### **4.2.5 Access to Experts**

Cambridge has deep expertise distributed across its departments, research institutes, and colleges. The University hosts world-leading scholars in areas such as manufacturing, innovation management, entrepreneurship, technology policy, and systems engineering. Many departments including the Institute for Manufacturing, the Judge Business School, the Maxwell Centre, engineering divisions, and policy research units – house academics whose research directly informs innovation ecosystems and technology translation.

These groups generate cutting-edge insights and guide the design of practical, evidence-based interventions for Cambridge’s startup system. Leveraging this broad institutional expertise ensures that the venture-creation model draws on diverse disciplinary perspectives and benefits from the University’s collective intellectual strength.

#### **4.2.6 Overall Assessment**

Cambridge’s educational assets form the foundation of a Training Layer within a broader startup production model. The key challenge is coordination, integration, and scale across existing programmes. By strengthening and aligning these assets, Cambridge can enable more students and researchers to graduate with the skills, confidence, and institutional support required to launch companies.

This can be done with:

- an Innovation Fellows PhD track
- interdisciplinary incubator-style courses
- structured alumni engagement systems
- expanded MBA and MET entrepreneurial pathways

### **4.3 Existing Research and Policy Infrastructure**

Cambridge suffers from invisible knowledge gaps. Many widely suspected ecosystem dynamics are difficult to confirm or measure because granular reporting is limited, especially regarding

negative data – information about failures, stalled ventures, or ideas that never progress beyond ideation. Innovation in Cambridge typically begins at the point of novel ideas, and many ideas with real potential never advance, despite the extraordinary intellectual capital in the city. These “*hidden failures*” are theoretically common but almost entirely unrecorded. Data is only collected once a founder interacts with formal infrastructure, and because Cambridge has limited support for pre-company stages, no such data is captured at earlier points.

Knowledge infrastructure gaps therefore include:

- absence of systematic failure capture
- lack of a searchable precedent database
- limited institutional memory of unsuccessful ventures
- no visibility into the knowledge conversion cycle (time from ideation to value added)
- no ecosystem-wide monitoring of resilience or other health metrics

Many of the indicators that are used in venture screening provide only a partial picture.

Common examples include:

- market size metrics (TAM / SAM / SOM)
- founding-team track record (proxied by publications, affiliations, patents)
- technical defensibility
- early revenue indicators (ARR / MRR; pilot contracts for deep tech)
- capital-efficiency ratios
- progress against sector-specific TRLs

These are valuable, but they do not capture the upstream and downstream dynamics of Cambridge’s innovation flow.

#### **4.3.1 Institutions Providing Research and Policy Insight**

Despite these gaps, Cambridge benefits from several institutions capable of generating data, policy insight, and ecosystem intelligence, all of which can be harnessed to identify and remedy invisible gaps:

##### **Bennett Institute for Public Policy**

- Conducts research on economic policy, including innovation, productivity, and regional development
- Can analyse how Cambridge’s innovation economy affects the region
- Advises on policy levers (R&D tax credits, visas, funding mechanisms) that influence the

startup landscape

- Leads the Cambridge branch of the UK Productivity Institute, whose work strongly aligns with upgrading Cambridge’s innovation model

### Departmental Entrepreneurship Hubs

- Some departments maintain hubs such as the IfM’s Centre for Technology Management or the Judge Business School’s entrepreneurship centre
- These hubs host workshops, publish case studies, and convene stakeholders
- The Judge’s Enterprise Tuesday is a university-wide forum facilitating learning from founders and operators

### Cambridge Cluster Data Tools

- Organisations such as Cambridge Ahead maintain dashboards, including the Data Hub, which compiles economic metrics (6).
- The Cambridge Cluster Map, hosted by the Cambridgeshire and Peterborough Combined Authority, maps company activity and growth patterns (7).
- Although these tools do not yet fully exploit big data or real-time analytics, they are an important starting point for building the Measurement Layer described earlier
- Local think tanks and analysts (including contributors to the Cambridge Independent Economic Review, and researchers at the Centre for Business Research) offer deep, contextual knowledge of Cambridge’s economic and innovation trends.

#### 4.3.2 Overall Assessment

Cambridge hosts a wide range of practical programmes for founders alongside deep intellectual and policy expertise capable of guiding system design, measurement, and optimisation. Few cities in the world contain such a concentrated body of researchers dedicated to studying innovation itself.

By using Cambridge as a *“living lab”* for innovation policy, these institutions can:

- inform the design of the startup creation model
- generate the missing data and metrics
- iterate and refine the system through evidence-based adjustments

This capacity positions Cambridge to build not just an innovation ecosystem, but a measurable and continuously improving innovation engine.

## 4.4 Existing Capital and Innovation Programmes

Earlier sections discussed formalised and structured models for innovation. Cambridge has launched a growing set of venture-building initiatives aligned with these principles. Historically, many of these efforts were disjointed and driven by individual departments or colleges. More recently, however, the University has introduced initiatives such as Founders@the University of Cambridge (75), the START Accelerator focused on venture science and deep-tech startups (76), and SYNC, a co-founder matching platform (77). Together, these initiatives create a university-driven pipeline from scientific talent → team formation → venture acceleration.

Additional programmes include the Trinity Bradfield Prize, which provides seed funding, mentoring, and membership at the Bradfield Centre in the Cambridge Science Park (72). Similar competitions and early-stage initiatives are organised by colleges, student societies, and incubators across the city.

For physical space, Cambridge hosts:

- ideaSpace (38)
- Barclays Eagle Labs (2)
- The Glasshouse (Innovate Cambridge) (39)
- A network of 40 business parks around Cambridge (74)

Multiple research departments operate highly active programmes, such as those within the Judge Business School, the Maxwell Centre, and the Institute for Manufacturing.

These include:

- EnterpriseTECH (14) and i-teams (37) – drawing on university talent to support spin-outs
- Accelerate Cambridge (13) – startup acceleration
- Impulse (78) – deep-tech venture acceleration

Human capital in Cambridge is vast, and programmes that leverage it are plentiful. Although Cambridge has a world-leading density of innovation capital and mechanisms, integration across these initiatives is limited. Tools for navigating the ecosystem are being developed – for example, the Cambridge Network maintains an open directory of venture-associated organisations (15) – but this is not yet operationalised into a unified, systematic pathway for founders.

### 4.4.1 Capital Providers and Support Programmes

Cambridge’s innovation ecosystem is supported by a set of maturing capital entities and structured programmes:

#### Cambridge Enterprise and University Seed Funds

- Cambridge Enterprise (CE) is the University’s tech transfer and venture arm.
- Operates the Combined University Seed Funds and manages proof-of-concept grants.
- Has helped launch hundreds of companies and reported the largest increase in spin-outs in 2024 among UK universities (9).
- Co-invests with VCs and runs an Enterprise Fund open to alumni investors.
- CE provides early-stage funding, IP support, legal resources, and founder mentorship – a core asset for university-affiliated ventures.

### **Cambridge Innovation Capital (CIC)**

- A venture fund linked to Cambridge, investing from seed to scale-up in deeptech and life sciences.
- Has backed major successes like CMR Surgical and Frontier.
- Its new Fund III commits at least £100M to university spin-outs (70).
- CIC provides a local source of scale-up capital, helping retain companies in Cambridge instead of pushing them to relocate.

### **Angel Investors and Networks**

- Cambridge hosts a strong angel community, including:
  - Cambridge Angels
  - Cambridge Capital Group
  - Sector-specific angel groups (e.g. life sciences)
  - Cambridge also benefits from London-based investors, with many VCs increasing their presence in Cambridge because of rising deal flow.

#### **4.4.2 Accelerators and Incubators**

Cambridge has several high-quality accelerators, including:

### **DeepTech Labs (DTL)**

- A Cambridge-based accelerator and VC fund focused on deep tech (25).
- Runs intensive 10-week cohorts, embedding founders in expert networks.
- Founded in partnership with the University, ARM, CIC, and others.
- Acts as a local startup foundry bridging prototype to product – a rare and valuable structure in the UK.

## Entrepreneur First (EF)

- A talent investor that identifies ambitious individuals and helps them form startups from the scratch, providing funding, mentorship, and a network (29).
- While EF is London-based, it attracts significant Cambridge talent.
- A pre-team, pre-idea accelerator; could be engaged for Cambridge-specific cohorts.

## Sector-Focused Incubators

- ideaSpace – general tech innovation
- Bio-Incubator at the Biomedical Campus – life sciences
- Accelerate@Babraham – biotech
- Additional incubator facilities across the science parks and research campuses

These provide specialised support, infrastructure, and community environments for targeted sectors.

### 4.4.3 Overall Assessment

Individually, Cambridge’s programmes are excellent. Collectively, they form a patchwork with gaps in integration, sequencing, and continuity.

The opportunity lies in systematising and connecting these assets so that startups can:

- emerge from college competitions
- join structured accelerators like DTL
- secure seed funding from CE, CIC, or angels
- access growth capital
- scale without leaving Cambridge

Scaling what already works – for instance, expanding DeepTech Labs or creating sector equivalents like a dedicated “*Biotech Labs*”, would increase conversion from idea to retained company. The assets are present; the next step is coordination and scale, enabling every promising venture to tap Cambridge’s full innovation stack with minimal friction.

## 4.5 Existing Scouting and Investment Efforts

- Proximo scout network
- Creator Fund in Cambridge
- College-based seed funding (small-scale)

In addition to formal programmes, Cambridge benefits from a number of grassroots, student-led, and micro-investment initiatives that scout early talent and ideas:

#### **4.5.1 Proximo Scout Network**

- A student-led scouting programme that identifies promising projects emerging from Cambridge labs and research groups (56).
- Scouts refer to ventures for funding, incubation, or accelerator pathways.
- Mirrors U.S. university models where students act as venture partners for VC firms, tapping into early-stage innovation before it becomes visible to institutions.

#### **4.5.2 Creator Fund (Student VC Presence)**

- A student-run VC fund active across the UK, with a significant presence in Cambridge (20).
- Provides students with hands-on experience in venture investing while offering small cheques to student-led startups.
- Complements larger Cambridge funds by capturing deals that are too early or too small for institutional investors.
- Reflects a growing entrepreneurial culture among students and contributes to early pipeline diversification.

#### **4.5.3 College-Based Seed Funds**

- A number of colleges have launched their own micro-seed initiatives
- Trinity College operates a donor-backed startup fund for Trinity founders.
- Colleges such as St John's and Jesus support entrepreneurship through societies, competitions, and grants.
- These are typically modest (< £20k per venture) but leverage college pride, alumni loyalty, and local networks.
- Coordinating or pooling these funds (e.g., into an inter-college seed vehicle) could increase their collective impact.

#### **4.5.4 Angel Investors and Mentor Capitalists**

- Cambridge has many independent angels – successful founders, wealthy alumni, and retired executives – who invest informally.
- Interviews on platforms like the Cambridge Tech Podcast suggest growing interest in international outreach, mentorship, and risk tolerance, especially from leaders such as Jim

Glasheen of Cambridge Enterprise Ventures.

- These individuals can champion cultural change by modelling:
- higher-risk investment behaviour
- modernised spin-out IP policies
- improved founder incentives

#### **4.5.5 Metrics in Use by Investors**

Traditional investor metrics include:

- market size
- team track record
- technical defensibility
- early revenue signals
- capital efficiency
- TRL-based progress

To mature the system, Cambridge could adopt more nuanced, ecosystem-aware metrics, such as:

- founder resilience
- a team's network centrality within the Cambridge cluster
- Cambridge-specific revenue pacing benchmarks for deeptech ventures

Developing Cambridge-specific benchmarks would help investors calibrate expectations to the realities of deeptech and science-led ventures.

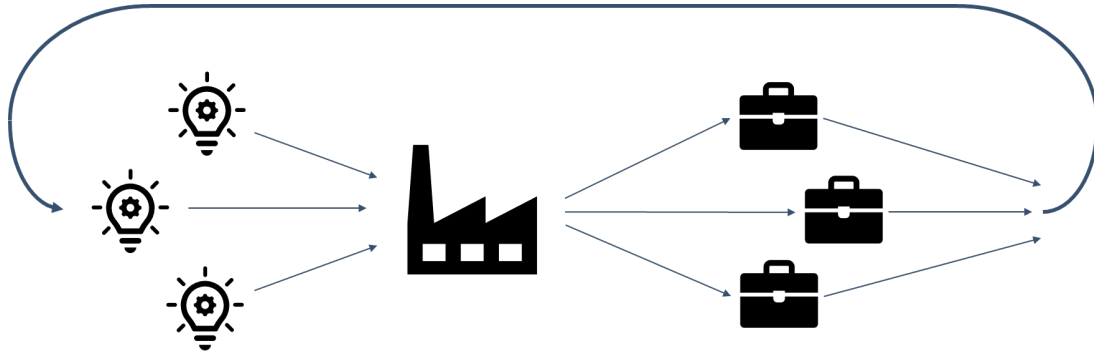
#### **4.5.6 Overall Assessment**

These grassroots and micro-initiatives demonstrate that Cambridge's innovation community is active, decentralised, and self-organising. A systemic solution should not stifle this bottom-up energy, but rather amplify, integrate, and coordinate it.

People tend to not create scout networks or micro-funds unless the ecosystem has gaps. Their emergence signals unmet needs and an appetite for more structure. By linking these efforts under a cohesive model – sharing information, co-investing, aligning deal flow, training scouts, and preventing duplication, Cambridge can ensure that no promising idea falls through the cracks, and that early pipeline scouting becomes a structured strength rather than an incidental advantage.

## 5 PROPOSED SOLUTIONS

### THE CAMBRIDGE STARTUP PRODUCTION MODEL



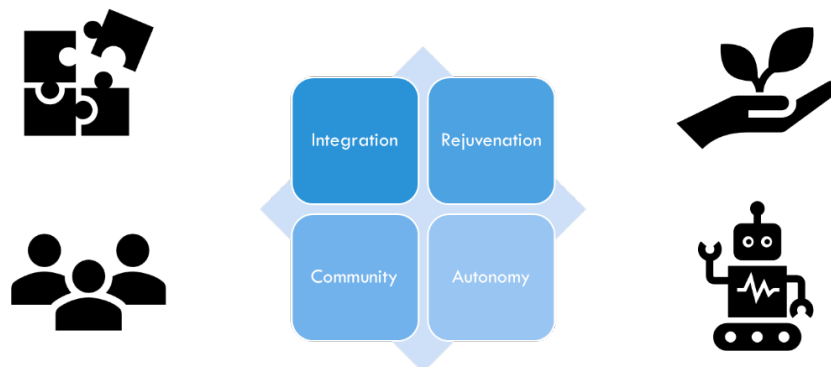
Cambridge possesses immense assets and a rich history of launching successful ventures. Its structure and resources are the foundation of its extraordinary scientific output. Rather than viewing Cambridge as facing a problem, we see an opportunity: to introduce operationalised infrastructure that complements its strengths.

We propose a **shared venture infrastructure model**, the Cambridge Startup Production Model. It is an engine for startup creation that supports business development without harming science. This model integrates data-driven systems thinking into Cambridge, enabling automation and coordination across the ecosystem.

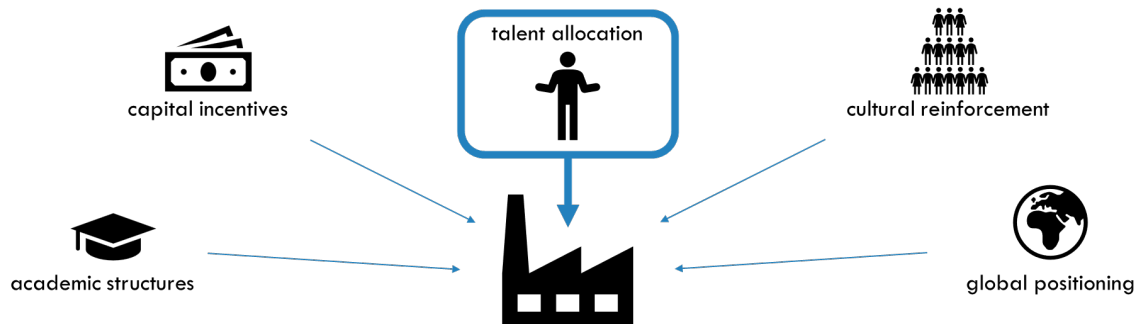
Key components include:

- A **common data layer** tracking startups and founder talent.
- A **city-wide incubator or portfolio strategy**.
- A **unified narrative** connecting colleges, labs, and investor operations.

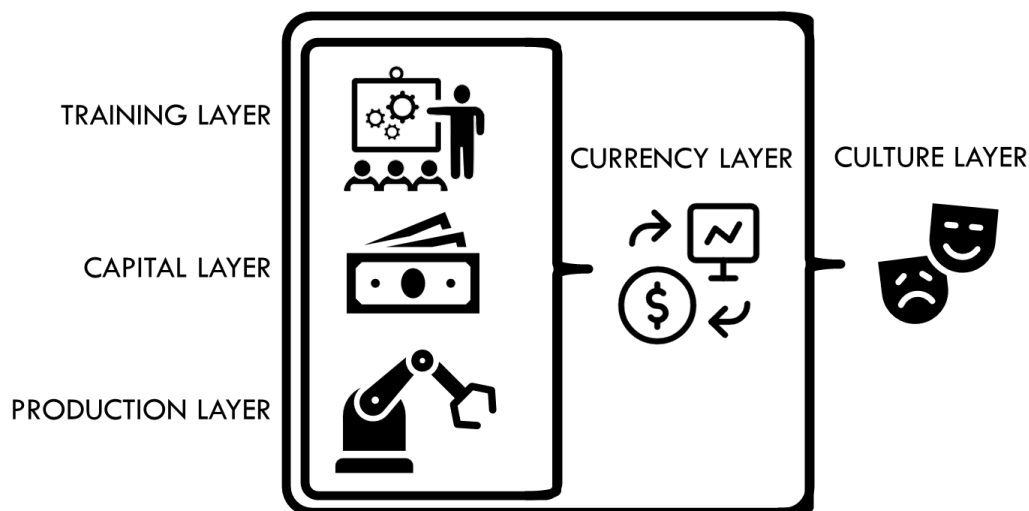
This approach reduces duplication, fills gaps, and allows coordinated portfolio construction at the ecosystem level. Investors currently manage their own deal flow and metrics; a shared database with detailed measures of venture performance provides transparency and enables optimal resource allocation. Shared knowledge about which spin-outs are succeeding or which sectors are saturated, allows Cambridge to learn from past failures and invest more strategically.



These interventions can be understood as filling the support of 4 core pillars: integration; rejuvenation; community; and autonomy. With these, they produce a self-maintaining innovation culture, built to amplify and leverage the existing ecosystem.



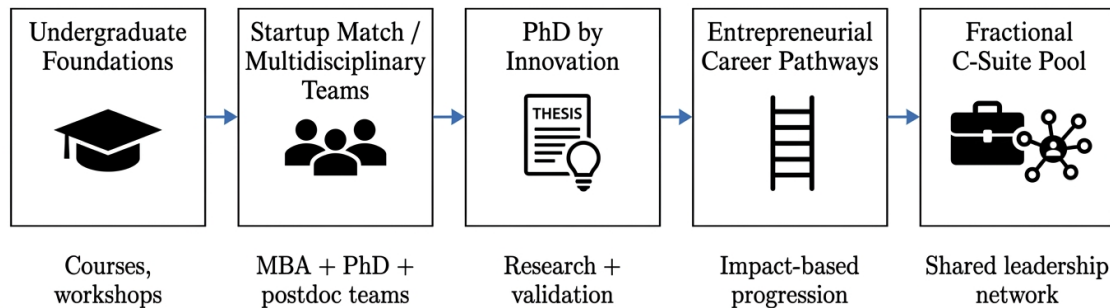
Functionally, they leverage and introduce integrated academic structures, capital incentives, cultural reinforcement, and global positioning into a single system with talent allocation. Interventions fall into either a training, capital, production, currency, or culture layer, creating a five-layer control system.



- Training layer interventions create an entrepreneurship education pipeline with many access points, integrating academic progress with entrepreneurship and constructing a sustainable innovator pool.
- Capital layer interventions use community-based financial mechanisms to promote growth and structure.
- Production layer interventions dynamically provide data, talent, and critical founder resources.
- Currency layer interventions are backed by the capital layer, leverage training layer talent, and govern the production layer by incentivising Cambridge innovation infrastructure reinvestment with a market.
- Culture layer interventions involve events, traditions, and practices that generate and

define a Cambridge-styled entrepreneurial culture, ensuring the resilience of other methods cheaply.

## 5.1 Training Layer



**Figure 4:** Training Layer Venture-Talent Pipeline

This layer organises Cambridge’s educational components for business creation. The objective is to build an educational pipeline that systematically produces entrepreneurially minded talent and matches people to ideas early. The Training Layer consolidates Cambridge’s vast educational resources into a coherent, structured programme for venture creation.

### 5.1.1 Undergraduate Entrepreneurship Foundations

At the earliest stage, every Cambridge undergraduate – or at least those in STEM and business, should have access to basic entrepreneurship education through electives or co-curricular workshops. Engineering students, for example, could be required to take a “*Concept to Commercialisation*” module in their final year.

Models for inspiration include:

- MIT’s Undergraduate Practice Opportunities Program (UPOP)
- Stanford’s entrepreneurship minor
- Cambridge’s existing Entrepreneurship MSt

The goal is to make entrepreneurial thinking part of the Cambridge mindset, not an extracurricular niche.

### 5.1.2 Integration with MBA and Business Talent

This applies to the educational foundation through structured teamwork. Drawing inspiration from EnterpriseTECH and i-teams, Cambridge can create a Startup Match programme in which multidisciplinary teams, working together for a semester on translating real Cambridge research into a prototype or commercial concept.

These may consist of, for example:

- 1 MBA student,

- 1 engineering PhD,
- and 1 life-sciences postdoc

This simulates the formation of founding teams. Studies show that diverse teams outperform homogeneous ones (34), and this model creates low-stakes simulations that can evolve into real companies when the chemistry is right.

### 5.1.3 PhD by Innovation

Cambridge should consider offering a formal doctoral pathway where the thesis is the creation of a viable business, analogous to a PhD by publication.

Features:

- Students split time between academic research and commercial validation
- Co-supervision by an academic and an industry mentor
- Coursework on entrepreneurship and innovation
- Dissertation outputs include a business plan or prototype, alongside academic work

Precedents:

- UK Engineering Doctorate (EngD) programmes
- ETH Zurich's Certificate of Advanced Study in Entrepreneurship, used by Pioneer Fellows

A **Cambridge PhD by Innovation** aligns academic incentives with venture creation, giving students permission to pursue commercialisation without jeopardising their academic standing. It rests on the principle that an effective startup relies on product-market fit, best determined through market application rather than abstract theorisation. While conceptual models of market needs are valuable, they do not guarantee that a novel innovation will embed successfully in the market.

At the start of a venture, both the target market segment and the solution are typically unknown—neither literature nor the founder provides complete guidance. Learning the appropriate pairing of market and solution typically follows three strategies:

1. **Market-first:** Identify the market segment first, then seek a solution. Useful for guidance, but it is uncertain whether a solution exists for the identified segment.
2. **Solution-first:** Develop the solution first, as in the scientific process, but risk it being inapplicable to any market niche.
3. **Simultaneous learning:** Apply available resources and knowledge to the market, iteratively adjusting based on feedback. Historically, this approach provides the most accurate insights into product-market fit and the path to implementation.

Externally, this iterative process may appear as repeated failure. However, each attempt

generates critical knowledge for the founder, representing expertise in a product-market niche. We argue that this iterative, evidence-driven process constitutes a discipline of study, justifying a **formal qualification**. It also reframes entrepreneurial failure as a natural part of product-market discovery, elevating the entrepreneurial journey to the same intellectual stature as scientific research. This approach challenges prejudices against entrepreneurship and those who have yet to achieve traditional “*success*,” emphasising study, experimentation, and learning as legitimate scholarly contributions.

#### 5.1.4 Entrepreneurial Career Pathways

Cambridge should establish formal academic career pathways where commercial impact is a primary promotion criterion.

This draws on:

- MIT Sloan’s Professor of the Practice roles (49)
- Monash University’s Academic Performance Framework emphasising translation and commercialisation (50)
- Recommendations of the UK Spin-out Review (71)
- Critiques from Times Higher Education (2025) noting the persistence of publication-centric evaluation

KPIs for such positions may include:

- number of ventures facilitated
- TRL advancement
- patent-to-product conversion
- founder retention in Cambridge
- time-to-seed reductions

These roles would help track negative data, strengthen the knowledge conversion cycle, and serve as role models for PhD by Innovation candidates.

#### 5.1.5 Fractional C-Suite Pool and Network

Training also includes building a structured mentor and leadership network, matching every new founding team with fractional C-suite professionals drawn from alumni and local industry.

Key features:

- Tokenised contribution model: C-suite members receive tokens for service, redeemable within Cambridge’s mutual-fund token system (explained later)
- Startups can “*purchase*” C-suite services with tokens, removing cash barriers

- Creates scalable access to leadership talent from day one
- Builds the talent infrastructure required for automated startup creation

### 5.1.6 Extending the Fractional C-Suite Model

The fractional C-suite network becomes both an educational platform and a talent-development pipeline.

Components include:

- structured rotation programmes
- cross-training between startups
- gamified incentives: reduced contribution mandates for hosting cross-trained members, token rewards for delivery

Possible specialisms emerging from this model:

- Technical + Regulatory + Commercial
- Academic + Operations + Finance
- Research + Design + Manufacturing
- Science + Policy + Investment

The model can function as a profit-generating educational product embedded in existing programmes such as the MBA, MET, or PhD (Innovation).

### 5.1.7 Outcome of the Training Layer

Implementing this layer would create a farm system for founders and startups, analogous to a university sports system cultivating athletes.

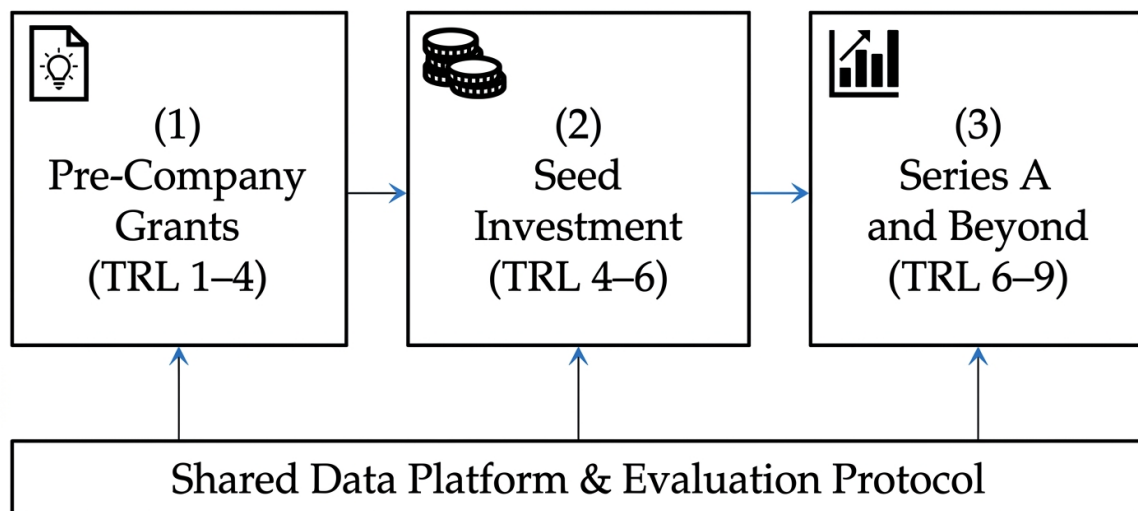
Expected benefits:

- A significantly larger pool of well-prepared founders
- More teams with both technical depth and entrepreneurial capability
- Pre-formed interdisciplinary teams ready for prototyping and venture acceleration
- Increased flow of credible startup ideas into the Capital and Production Layers
- Cultural transformation: when hundreds of students work on startup ideas as part of their curriculum, entrepreneurship becomes normalised across Cambridge

The Training Layer lays the foundation for a systematic, repeatable, and scalable startup creation engine.

## 5.2 CAPITAL LAYER

To finance the services that enable ventures to satisfy the required criteria and contribute to Cambridge’s growth, a **community funding mechanism** can be introduced. This layer provides mechanisms to fund ventures within the ecosystem, with an emphasis on supporting real-growth businesses. Funding models should be designed both to stimulate rapid revenue generation and to ensure long-term ownership retention.



**Figure 5:** Capital Layer Tiered Funding Progression

This mechanism also forms a key pillar through which further interventions to scale startup creation in Cambridge can be financed and managed. The goal is to create a **closed entrepreneurial loop** within Cambridge, where successful ventures help fund the next generation of startups, accelerating the rate at which novel innovation emerges from Cambridge and reaches the global market.

### 5.2.1

#### 5.2.2 Cambridge Community Fund (1–1–1 model)

We propose a Cambridge Community Fund, adapting the proven model pioneered by Salesforce through its 1-1-1 framework, which commits 1% of equity, time, and product to community benefit (57). This model has been adopted by over 18,000 companies through the Pledge 1% initiative. In our proposal, the model is adapted into a revenue-based mechanism designed for ecosystem retention rather than philanthropy.

The structure—£1 membership, a 1% revenue contribution, and a one-year delay before contributions begin draws on **revenue-based financing approaches** used by cooperative accelerators such as Start.coop (61). These models use quarterly revenue payback rather than equity dilution, allowing founders to retain ownership while contributing to ecosystem sustainability. The one-year delay reflects the “*runway protection*” principle, ensuring ventures reach early stability before making contributions.

To address common criticisms of flat-rate contribution systems, the model introduces floors and caps: companies generating less than £10k in revenue are exempt, while contributions from high-growth ventures are capped at £1M. This approach mirrors progressive structures seen in programmes such as the Yozma Programme, where government returns were reinvested into new venture funds rather than extracted as profit. That programme is widely credited with increasing venture capital investment in Israel from roughly \$5 million to \$3.3 billion within a decade.

Unlike equity-based community funding schemes, a revenue-based contribution preserves founder control while creating a self-sustaining funding cycle for ecosystem services, enabling Cambridge to continuously finance the infrastructure that supports startup creation.

**Venture Cooperatives** The fund is complemented by the concept of a venture cooperative. Under this structure, core technology is held cooperatively, while individual ventures license it and contribute profits back into an evergreen fund, allowing capital to be continuously recycled into new ventures. This approach adapts proven cooperative models to the Cambridge ecosystem.

For example, Start.coop demonstrates how cooperative ownership can function in technology startups: participating companies graduate into a shared cooperative structure that distributes ownership among workers, investors, and portfolio companies (61). Similarly, European startup cooperative frameworks allow flexible ownership arrangements in which new investors can join without requiring notarial restructuring. These structures enable deals to be configured flexibly across different growth phases, while members collectively determine profit-sharing arrangements and voting rights.

Together, these models illustrate how cooperative ownership can support venture creation while maintaining founder participation and long-term ecosystem alignment.

**Evergreen Capital** We use the concept of an **evergreen fund** to define how business outputs from participating firms contribute to financing the ecosystem beyond traditional returns models. Unlike closed-end venture funds, evergreen funds remain open indefinitely. When an exit occurs, capital is automatically recycled into new investments, creating a perpetual investment cycle without requiring continuous inflows of outside capital.

This model draws on established precedents. For example, Startup SG Equity in Singapore provides a clear government co-investment framework (28). The programme has invested nearly S\$3 billion across more than 330 startups, with over S\$2.5 billion sourced from private-sector partners (59). For deep-tech ventures in particular, the Singapore government often provides up to 70% of early-stage funding before transitioning to a 1:1 matching model with private investors (59), demonstrating how public capital can successfully leverage private investment at scale.

Cambridge could pursue a localised analogue through existing UK mechanisms. Programmes

such as Innovate UK’s Investor Partnerships already provide matched private-equity investment, while regional innovation Launchpads have received £75 million to accelerate local economies through targeted SME support. By integrating these tools within an evergreen fund structure, Cambridge could align individual incentives with collective ecosystem growth while avoiding some of the extractive dynamics associated with traditional venture capital.

**Finance via Tutoring Ecosystem** Evergreen funds typically rely on ongoing funding sources rather than periodic capital raises, allowing them to avoid traditional fundraising cycles while continuously reinvesting returns. In this model, the tutoring system developed within the training layer serves not only internal educational purposes but can also operate as a revenue-generating service.

Cambridge’s existing educational ecosystem—spanning tutoring, training, and skills development, could therefore channel a portion of its profits into the venture cooperative, helping to finance the ecosystem that supports it. In this way, educational activity contributes directly to venture creation, creating a self-reinforcing loop in which learning, enterprise, and investment sustain one another.

### 5.2.3 Financial Infrastructure

- Cambridge Bond (IP-backed)
- Cambridge Token (governance/utility)
- Cambridge Stock Exchange (venture progression tiers)
- Cambridge Real-estate financed fund.

**Stock Exchange** The UK’s new Private Intermittent Securities and Capital Exchange System (PISCES) framework provides a novel pathway for private companies to access intermittent liquidity (32). In August 2025, the London Stock Exchange became the first operator approved by the Financial Conduct Authority to run this system, allowing private firms to conduct periodic trading auctions while remaining private. The framework enables startups to test the waters of public markets while retaining control over when and how their shares trade. The HM Treasury has also confirmed that PISCES transactions will be exempt from stamp duty, further lowering participation barriers.

A Cambridge-specific trading hub could act as an early adopter within this framework, creating structured progression pathways for ventures within the local ecosystem. Venture exchanges operating under such a structure offer transparent and regulated environments with enhanced liquidity and investor protections, and can be designed to focus on companies emerging from a particular geographic innovation cluster.

One advantage of this approach is that a Cambridge venture exchange would establish a central trading venue for local startups, generating a shared data source and clearer visibility into the

region’s venture dynamics. Participation standards could also formalise progression pathways, guiding ventures through defined innovation stages and contributing to a more organised and scalable startup creation environment.

**Cambridge IP Bond** The creation of a venture exchange and the data used to onboard companies to it, could provide sufficient insight to evaluate the performance of Cambridge’s intellectual property ecosystem in a dynamic and secure way. This visibility opens the possibility of introducing a Cambridge Bond, an intellectual-property-backed security structured with risk-tiered tranches.

Such instruments have clear precedents in financial markets. In 1997, David Bowie pioneered IP securitisation by raising \$55 million through bonds backed by future royalty revenues (19). Similar structures were later used by Coca-Cola, which raised \$1.5 billion in 2001 (33), and The Walt Disney Company, which raised \$1.7 billion in 2007 (33). In these arrangements, intellectual property assets are bundled and sold to investors as securities representing future revenue streams such as royalties, licensing fees, and other forms of IP income.

A comparable structure could be applied to Cambridge’s concentrated portfolio of research-derived intellectual property, potentially supported by government guarantees for senior tranches. Institutional capital could also play a role. For example, the British Business Bank’s British Growth Partnership is targeting £200 million for its first fund, while 17 of the UK’s largest workplace pension providers have committed to investing at least 5% of default funds into UK productive assets—an initiative expected to unlock more than £50 billion by the end of the decade. The Bank’s Venture Link programme further aims to help pension funds develop venture investment strategies and reduce barriers to allocating capital toward UK science and innovation.

Cambridge could position itself as a regional demonstration case where these mechanisms converge:

- Private Intermittent Securities and Capital Exchange System (PISCES) for liquidity,
- IP-backed bonds for predictable funding streams, and
- pension capital for long-term scale.

Together, these mechanisms would create an **integrated capital layer** capable of sustaining venture growth while keeping value generated from Cambridge innovation anchored locally.

#### 5.2.4 Investment Decision Reform

Considering these financial interventions, the next question is how they can be implemented at scale and how the resulting capital should be deployed. Cambridge contains an unusually high concentration of world-leading technical and business expertise across its colleges, departments, and research institutes. If properly leveraged, this expertise could be operationalised to support investment decision-making, distributing evaluation capacity across the ecosystem and

increasing the effectiveness of the financial mechanisms described above.

The core intervention is relatively straightforward. Lecturers, research staff, and advanced students possess both membership within the Cambridge community and domain-specific expertise that external investors often cannot replicate. By equipping these actors with standardised evaluation frameworks and connecting them through shared data infrastructure, Cambridge could create a distributed system for venture assessment.

Such a system would accelerate investment cycles, improve the accuracy of technical evaluations, broaden access to capital, and reduce the distorting influence of personal networks on funding outcomes. In effect, Cambridge's academic community would function as a collective intelligence layer supporting venture formation and investment across the ecosystem.

**Cambridge Evaluation Protocol** Research funding worldwide relies heavily on peer review, yet evidence suggests these processes are susceptible to biases and inconsistencies, with studies showing weak predictive power for future research performance (73). Conventional venture capital faces a parallel problem: deals are often screened by junior associates lacking technical depth, which is particularly problematic for deep-tech ventures. These ventures require 35% more time and 48% more capital than traditional startups, making evaluation by technically qualified experts essential (31).

The Cambridge Evaluation Protocol addresses both issues through a unified mechanism. Pre-company research projects apply for milestone-based grants anchored to Technology Readiness Levels, which are evaluated by a consortium of Cambridge-based domain experts, including professors, senior researchers, and technical specialists. Projects that achieve their milestones automatically qualify for seed investment consideration through the Cambridge Direct Investment Protocol, where a similarly qualified consortium of investors and experts evaluates ventures using the same platform and accumulated assessment history.

By combining technical evaluation with business-expert screening, this system enables more reproducible assessments, standardized terms, and smaller ticket sizes across a larger number of investments. The investor syndicate allows for deeper due diligence than individual investors could provide while pooling capital to reach appropriate round sizes (68). Special Purpose Vehicles can simplify cap table management when multiple syndicate members participate in a single deal.

Integration with alumni and corporate channels extends the system's reach and complements the Community Fund discussed earlier, potentially functioning as a sidecar fund. Alumni venture funds have demonstrated the effectiveness of this approach: for example, Princeton Alumni Entrepreneurs Fund leveraged \$2.2 million into nearly \$160 million in outside investment (21). Corporate venture partners, including ARM, AstraZeneca, and Microsoft Research, similarly route investments through the platform, gaining early visibility into relevant technologies while providing ventures with validation and strategic pathways.

### 5.2.5 Shared Data Platform

The shared data infrastructure underpinning this consolidates expertise and eliminates redundant assessment. Standardised due diligence questionnaires have been industry standard in institutional investment for over 20 years, enabling like-for-like comparison while reducing workload. Digital platforms demonstrate the efficiency gains possible when data collection, tracking, and analysis are centralised (69).

For Cambridge, ventures would submit a standardised data package covering technical descriptions, Technology Readiness Level (TRL) self-assessments, team composition, IP position, and milestone roadmaps. This package would serve multiple purposes—grant applications, investment pitches, and consortium circulation, without requiring reformatting. The platform would route applications to evaluators matched by research area and declared interests, and potentially by empirically measured expertise. Evaluations would accumulate as structured ratings accompanied by narrative commentary, creating an institutional memory: evaluators could see how previous similar ventures progressed, which technical risks materialised, which milestone frameworks were realistic, and learn to spot early warning signs. Integration with the Cambridge Failure Archive further enhances this learning.

By standardising criteria and accumulating assessments, the platform reduces the influence of social networks on funding outcomes. Ventures with strong technical merit but weaker social connections would receive evaluations equivalent to well-networked but technically weaker competitors. This aligns with research recommendations for meritocratic funding based on proposal quality rather than institutional affiliation (79).

### 5.2.6 Tiered Funding Progression

All of these mechanisms can be integrated into a broader, tiered funding system structured around empirically defined and reproducibly reported Technology Readiness Levels (TRLs).

We define it here:

- Pre-Company Grants (TRL 1-4): Milestone-based grants for projects emerging from the Training Layer or research labs. Evaluation by matched domain experts. Achievement triggers qualification for seed consideration.
- Seed Investment (TRL 4-6): Community Fund and consortium angels provide capital, with evaluation substantially complete from grant stage. Protocol terms apply: standardised documentation, milestone-based tranches.
- Series A and Beyond (TRL 6-9): Cambridge Innovation Capital and partner VCs lead larger rounds, drawing on consortium relationships and platform data. Quarterly meetings surface co-investment opportunities ensuring ventures find lead investors despite thesis-mismatch at any individual fund.

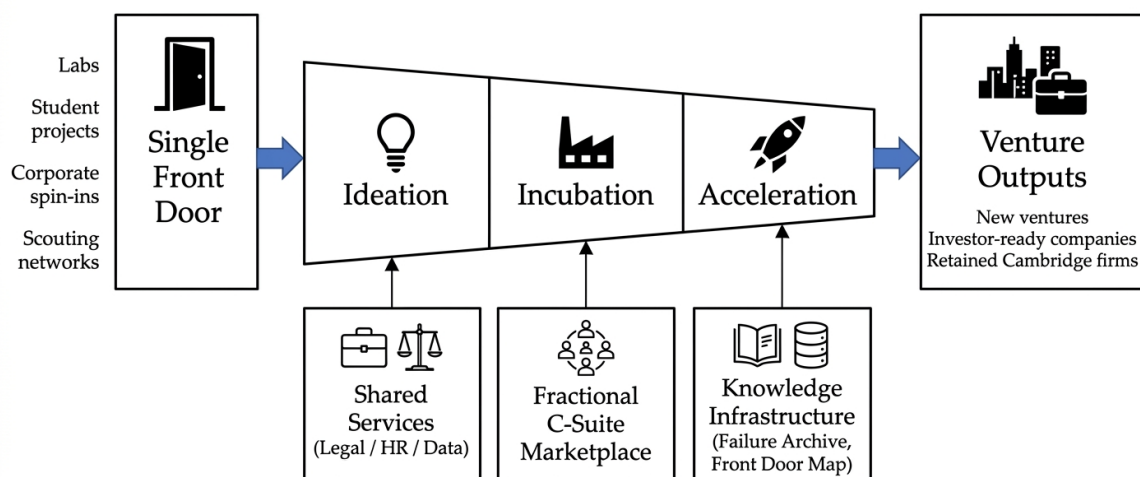
Right-of-first-refusal mechanisms ensure continuity and alignment within the ecosystem. For

example, the Community Fund would have first access to grant-stage companies that achieve their milestones, while consortium members would have first access to seed-stage ventures raising subsequent rounds.

In addition to financial advantages, greater involvement of university members in evaluation generates cultural benefits: it fosters closer relationships with ventures, exposes participants to commercialisation thinking, and strengthens alumni engagement. These benefits are discussed in more detail elsewhere in this document.

### 5.3 PRODUCTION LAYER

This layer constructs a formalised pipeline that draws ventures into a shared “*venture foundry*.” It provides coordinated services across the ecosystem while aligning its focus with the expectations of international investors. Sectors are subdivided into broad categories, and ventures enter through a single front door, benefiting from dynamic mapping of relationships and resources, a shared C-suite marketplace, and standardised venture creation processes.



**Figure 6:** Production Layer Venture Foundry

The Production Layer is the heart of the model—where all inputs, including talent, ideas, and capital, converge to produce new companies in a repeatable and scalable way. It transforms the Cambridge ecosystem from a collection of disconnected capabilities into a **structured engine for venture creation**.

#### 5.3.1 The Venture Foundry System

This creates a formalised venture pipeline (“*venture foundry*”) that draws in ideas from across Cambridge and systematically turns them into startups through shared infrastructure and services.

**Cambridge Venture Foundry (CVF)** Establish a central entity, it could be a physical hub or a coordinated virtual program. This functions as Cambridge’s startup assembly line. Ideas enter at one end (it can be sourced from labs, student projects, corporate spin-ins, or other

initiatives), and progress through stages of refinement, team formation, product development, and company launch.

The Foundry would have dedicated staff like entrepreneurial leads, engineers, and business analysts, who work alongside founders to guide the process. It can be thought of as a hybrid of an accelerator, a startup studio and a co-working hub, combining structured support, operational capacity, and community integration to produce ventures in a repeatable, scalable way.

**Shared Services Platform** This can manifest as shared services that are expensive or inefficient for each startup to do alone. Examples include:

- **Legal support** – standard incorporation documents and IP agreements, potentially coordinated with Cambridge Enterprise to simplify spin-out terms.
- **Accounting and HR** – a central team managing payroll and basic financial tasks for multiple small startups, removing the need for each to hire separate staff.
- **Shared data resources** – for instance, a CRM of industry contacts accessible to all startups, or a database of experimental results (with appropriate NDAs) for ventures working on overlapping domains, such as gene sequencing.

By pooling these resources, startups can **accelerate development and reduce operational overhead**. Station F in Paris provides a compelling example: housing 1,000 startups under one roof, with access to consulting, fab labs, and shared infrastructure, has made it a magnet for entrepreneurial activity (64). Cambridge could replicate this model at scale, adapted to its unique academic and research ecosystem.

**Venture Tiers and Gates** Much like academic programs progress year by year, the Cambridge Venture Foundry will implement tiered stages for startups:

- **Ideation Stage** – ventures are exploring ideas but not yet incorporated; many enter this stage and receive light support.
- **Incubation Stage** – the most promising ideas are incorporated and receive initial funding and moderate support.
- **Acceleration Stage** – a select few receive significant funding and intensive support to scale.

These stages can be mapped to Technology Readiness Levels (TRLs), previously assessed through the evaluation protocols, simplifying stage placement and progress tracking. Ventures that do not advance should be recycled into alternative initiatives, directing founders toward more productive paths for the ecosystem.

**Physical Hub and Innovation Districts** While the Production Layer concept does not require a single building, having a physical nexus can greatly enhance its effectiveness. This proposal could include developing a Cambridge Innovation District as the headquarters of the venture foundry. Co-locating startups increases the likelihood of serendipitous collaboration, both informal and structured, while simplifying the operational coordination of shared services and ecosystem activities.

A clear precedent is Kendall Square, Cambridge, MA, where the mix of startups, corporate labs, and academic offices created a globally recognised innovation powerhouse (60). Cambridge can aim to replicate this by developing a high-density innovation quarter, similar to the evolving Cambridge Biomedical Campus for life sciences but focused on technology and venture infrastructure. By encouraging startups, incubators, and VCs to cluster in one or two key areas, the city can create a hub where capital, talent, and ideas converge.

A central location also benefits visiting investors and partners, serving as a “*one-stop shop*” to experience Cambridge innovation in action. This physical concentration complements the broader Production Layer, combining repeatable venture creation processes with the advantages of proximity and visibility.

**Global Ties and Expansion** The Foundry should include programs that connect Cambridge startups with global markets without requiring relocation. For example, partnerships with accelerators in the US or Asia could offer 2-week immersion programs, or overseas VC firms could be invited to host demo days in Cambridge. The goal is to avoid the common pattern of startups feeling compelled to move to Silicon Valley to raise Series B funding—instead, Cambridge brings Silicon Valley to them, virtually or periodically.

Recent data underscores Cambridge’s strength: 41% of Cambridge startups reach Series A, compared with 40% in Silicon Valley (23), making it a compelling metric to showcase Cambridge as a high-growth, investable ecosystem. By leveraging such data, the Foundry can actively market Cambridge ventures internationally, attracting investor interest while keeping companies rooted in the local ecosystem.

### 5.3.2 Knowledge Infrastructure

- Cambridge Failure Archive
- Precedent database
- AI-powered failure prediction

To guide operations at scale, clear guides to the Cambridge landscape and insights into it are necessary. Research into Cambridge as a topic itself is important as we move beyond this report.

**Failure Archive** A proposed intervention is the Cambridge Failure Archive. It would document all venture failures with structured post-mortems, create a searchable database of

failure patterns, provide AI-based failure prediction for new ventures, offer confidential consultations based on comparable cases, and recognise intelligent failures annually.

**Dynamic Front Door to Cambridge** A dynamic “*front door*” to Cambridge should be created to help external and local participants navigate the ecosystem. It would map cofounder relationships, incubators, accelerators, research departments, private enterprises, laboratories, colleges, institutions, and the city’s technology hubs and business parks. This would reduce barriers for attracting both businesses and capital into the Cambridge ecosystem. It would also enable statistical identification of gaps in growth or ecosystemic richness, allowing for targeted policy intervention. Such analysis could be conducted using ecosystem-mapping techniques and Ricci curvature analysis.

**Lab-Space Allocation Mechanism** A plausible solution to the challenge of lab space utilization is a centralised lab-space matching platform operating as a marketplace. Founders, researchers, and companies would list their requirements such as equipment needs, biosafety levels, collaboration preferences, and budget—while labs would specify available capacity, pricing, and expertise. A matching algorithm would then optimise assignments based on compatibility, cost, and broader ecosystem priorities.

Additional interventions could enhance engagement and efficiency, including investor-badge system for verified funders, monthly “*lab crawls*” as structured tours, reverse-pitching whereby labs present their problems to investors, and an equipment time-sharing marketplace.

**Metric Development** Cambridge should invest in metrics specifically designed to upgrade the ecosystem, beyond standard VC performance metrics. Cambridge needs KPIs to measure the impact of interventions. These must be adopted at multiple levels and actively recorded, requiring a systematic data-collection paradigm as part of the startup-systematisation effort.

Suitable KPIs include:

- Retention rate of ventures in Cambridge
- Time-to-seed reduction
- Time-to-first-customer acceleration
- Cost-to-operate per founder
- Conversion rate from research to company
- Scale-up success-rate improvement

However, new ecosystem-upgrading metrics should also be used:

- Founder retention rate (% still in Cambridge after 3 years)
- Revenue per employee

- Time to first customer
- Number of customers who later buy from other Cambridge startups
- Number of customers sourced from Cambridge
- Cost per job created

These metrics can be reported through a monthly index focused on community richness rather than valuations. The aim is to identify areas where Cambridge has world-leading expertise (such as AI, genomics, quantum technologies, and material science) and align venture creation efforts with these strengths.

Strengthening the bridges between academia and enterprise is key. This requires integrating academic incentives with startup creation, for example:

- Allowing **entrepreneurial impact** to count toward academic promotions along an innovation-driven pathway.
- Creating **joint appointments** for practitioner-entrepreneurs who combine academic and commercial roles.

By making these connections explicit, Cambridge can ensure that venture creation complements its academic mission while embedding entrepreneurship into the culture of research and teaching.

**The Cambridge Intra-Cultural Exchange Protocol** To target the absence of stable productive weak ties between academic and commercial actors, we propose leveraging existing Cambridge entrepreneurial meetings through the Cambridge Intra-Cultural Exchange. This program rewards the introduction of new human capital into the network by granting “*connection credits*” to members who bring impactful individuals into the ecosystem.

Specifically, introductions are tracked, and community members log whether these connections lead to further introductions or collaborations. In return, participants earn Cambridge community currency, incentivising both data enrichment and service in integrating the network.

By mandating lightweight, regular meetings and measuring success through the creation of subsequent introductions rather than immediate collaborations, the protocol strengthens cross-network connectivity without imposing heavy obligations. The introduction of connection credits embeds the creation of social capital into the economic logic of the ecosystem, aligning individual incentives with the growth and integration of Cambridge’s broader innovation network.

To participate:

- Participants commit to at least three meetings per quarter
- Success is measured by subsequent introductions rather than direct collaboration
- Participants earn “*connection credits*,” redeemable for access to facilities or events, or

convertible into the Cambridge community currency

**Direct Market Visibility and Research–Commercial Alignment** It is important for Cambridge to have clearer visibility into the technologies and scientific outputs that the global customer base is actively seeking. Currently, this information is received indirectly through funding calls and investor theses, which are often lagging, non-specific, and misaligned with market demand. The result is technologies that are sometimes poorly adopted.

If direct calls for specific technologies were communicated to the scientific community, researchers could develop solutions already aligned with market needs, effectively bypassing the early, high-risk phase of determining product–market fit. This approach reduces failure points and shortens the time to commercialisation.

Early exposure to the market also provides ventures with access to customers, teams, and lab space sooner in the development process. This aligns with research by Moro-Visconti, which demonstrates that the value of know-how often exceeds the value of capital during significant portions of a company’s lifecycle (51).

Such a system could operate through KPI thresholds, including:

Customer evidence

- Number of customers
- Number of references
- Documented use cases

Unit economics

- Cost per acquisition
- Lifetime product value

Team

- Quantified and ranked team composition

Cambridge commitment could be demonstrated by:

- A local hiring plan
- Identification of local suppliers
- Metrics supplied by the Global Cambridge Initiative

**Cambridge Research–Commercial Synthesis (RCS)** These KPIs are propositional; further research is needed to determine the most appropriate metrics and corresponding thresholds. The Bennett Institute and the Productivity Unit offer a strong research base within Cambridge to support this, and an internal unit (the RCS Entity) could be established to

bridge academic and commercial domains.

Characteristics of the RCS Entity:

- Dual reporting to the university and a board
- Academic credit for commercial milestones
- Protected time for both research and development
- Shared IP ownership with clear conversion paths
- Pension and benefit continuity

### 5.3.3 Operational Talent Infrastructure

- Fractional C-suite marketplace
- Human Capital Credits
- Peer-to-peer bidding
- Rotations, cross-training, gamification

**Fractional C-Suite Marketplace and Human Capital Credits** Cambridge should implement a system of fractional C-suite officer assignments to ensure that advanced technological talent is efficiently matched with the right human expertise. The system’s design should prioritize ease, speed, minimal social friction, openness, and fairness, enabling experts to support startups from inception through scale-up before transitioning responsibilities to those better suited for later-stage growth.

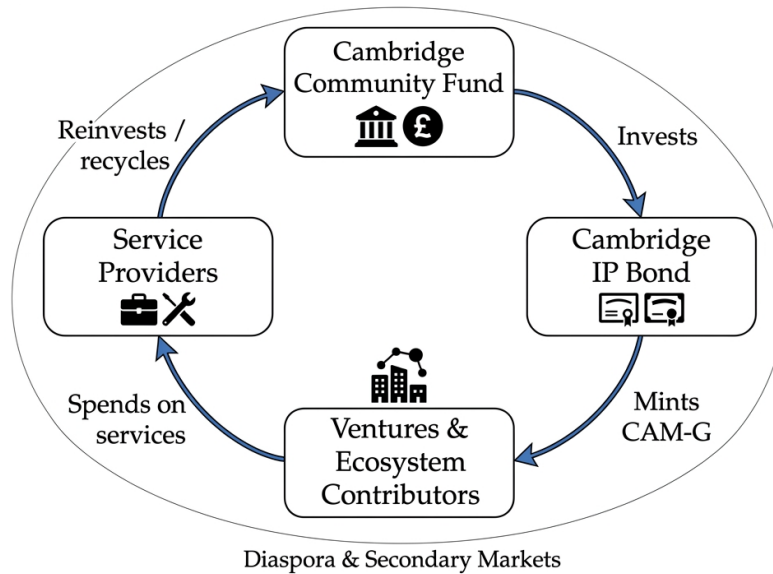
At the core of this system would be a marketplace for C-suite roles, allowing peer-to-peer bidding between startups and expert cofounders. To attract and retain top-tier startup talent, a token-based ownership system could be introduced, potentially leveraging blockchain technology. Under this model, a fractional C-suite’s objective extends beyond the success of individual startups; it becomes aligned with upgrading Cambridge’s broader ecosystem, as returns are linked to the overall value of the currency and the wider community. This approach capitalizes on Cambridge’s abundant and world-leading human capital.

The system could operate by exchanging hours of expert time for equity or revenue stakes in a community-wide fund, conceptualized as “*human capital credits*.” Founders would earn tokens based on the successful delivery of business functions, either through ownership stakes in ventures (e.g., Series A-level convertible positions feeding an intermediary market within the city) or through returns from the community fund. Allocations would be determined via a public bounty board, where founders post specific business functions and experts bid competitively to fulfil them, ensuring fairness and transparency.

This model positions Cambridge as a dynamic, self-reinforcing innovation ecosystem, where talent is efficiently mobilized, value is captured communally, and both startups and the city

benefit from aligned incentives.

## 5.4 CURRENCY LAYER



**Figure 7:** Currency Layer Ecosystem Token

This combines all principles of ecosystem building, capital management, venture production, with an emphasis on human capital dynamics and ecosystem insight capture. We propose that an independent token system, integrated with other financial interventions, be how automation is truly achieved.

### 5.4.1 The Cambridge Innovation Ecosystem Token

To implement this, such a token can follow the following principles:

- Issuance tied to Fund contribution via IP Bond investment: tokens only enter circulation through productive contribution to Cambridge, never through external purchase.
- Asymmetric vesting drives circulation: ventures must spend (services) because they cannot liquidate before the completion of a vesting period; support actors can liquidate sooner, so spending is rational for ventures and receiving is valuable for providers.
- Appreciation absorbed by Fund's growing IP Bond position: growth is pegged to the Cambridge IP bond, and we issue tokens commensurately so that new entrants aren't priced out. This is because the Fund's capacity to issue grows with the underlying asset.
- Trading permitted and encouraged: speculation performs price discovery and funds ecosystem development through transactions.
- Service pricing via market discovery: accessibility achieved through abundance as ecosystem matures.

- Diaspora dividend: tokens allocated to venture that have since expanded internationally retain Cambridge engagement, operationalising the "projection not extraction" goal

We have, in a sense, outlined a twin portfolio – one of venture support service providers, and one of ventures who have access to these services. This constitutes a fund, taking on the flavour of an “*Entrepreneurial College*”. Given the twin portfolio structure outlined above and a portfolio of ventures and business support actors operating within Cambridge, a token mechanism can formalise the exchange relationships between these portfolios while aligning all participants with the growth of the Cambridge ecosystem.

Using the Cambridge IP Bond as a value anchor, and the Cambridge Community Fund as the issuance mechanism, this token system simultaneously achieves three objectives:

- it provides ventures with access to business support services
- it injects capital into ecosystem builders
- it creates speculative alignment with Cambridge's aggregate commercialisation success.

**The Issuance Mechanism** The Cambridge Community Fund, structured according to the 1-1-1 model described earlier, serves as the token issuance gateway. When ventures contribute to the Fund whether through initial buy-in, revenue-based contributions post-threshold, or matched institutional capital the Fund invests these contributions into the Cambridge IP Bond via an automated market maker function. This investment action triggers token issuance: the contributor receives Cambridge Growth Tokens (CAM-G) proportional to their contribution, minted at the current position on the bonding curve. The bonding curve ensures that token price rises algorithmically with cumulative investment into the IP Bond, rewarding earlier contributors while maintaining continuous liquidity without traditional market makers. Critically, only Cambridge ecosystem investors, the fund, and its constituent structures can perform this exchange; CAM-G cannot be acquired except through contribution to Cambridge's productive capacity. This restriction ensures that token holders are, by definition, participants in Cambridge's growth rather than external speculators.

**The Circulation Mechanism: Asymmetric Vesting** To ensure tokens circulate productively rather than being hoarded, the system imposes asymmetric vesting constraints across portfolio participants. Ventures receiving CAM-G face extended vesting periods (18-36 months) during which tokens cannot be exchanged for fiat currency. Business support actors the fractional C-suite marketplace participants, training providers, laboratory operators, and advisory services described in the Production Layer face significantly lighter vesting requirements (3-6 months), enabling them to liquidate tokens and reinvest in operational improvements. This asymmetry creates a clear spending incentive: ventures holding illiquid tokens can either retain them as appreciating but unusable assets, or deploy them to purchase services from support actors who can actually realise the value. Since early-stage ventures typically achieve higher returns on invested capability than on passive asset appreciation,

spending dominates holding as the rational strategy. The support actors, upon receiving tokens, can partially liquidate to upgrade their operations better equipment, expanded capacity, enhanced expertise which improves service quality, which improves venture outcomes, which improves IP Bond performance, which appreciates the token. Value does not leak from the system; it transforms into ecosystem capability and returns via bond appreciation.

**The Appreciation Dynamic and New Entrant Accessibility** A natural concern with any appreciating token is that rising prices exclude new participants. However, the structure resolves this through the fund's expanding issuance capacity. As Cambridge ventures succeed and the IP Bond appreciates, the Fund's underlying asset base grows proportionally. This expanded base supports proportionally expanded token issuance to new entrants at accessible effective prices. The appreciation cost is absorbed by the fund's growing IP Bond position, not passed to newcomers. In effect, Cambridge itself through the fund acting as ecosystem investor underwrites the accessibility of future participants. This is analogous to equity dilution logic applied at ecosystem scale: existing holders see their percentage ownership dilute as new tokens issue, but absolute value grows because the underlying asset (aggregate Cambridge commercialisation success) expands faster than issuance. The fund's dual position as IP Bond holder and token issuer makes this sustainable: appreciation in one capacity funds accessibility in the other.

**Trading and Speculation** CAM-G holders can trade tokens on secondary markets, speculating on Cambridge's aggregate commercialisation trajectory. This speculation is not merely permitted but encouraged, as it performs a price discovery function and creates liquidity for participants who have completed vesting. Importantly, trading activity itself contributes to ecosystem development: following the Augmented Bonding Curve model, a percentage of each transaction flows into a common pool of funding ecosystem development grants, Cambridge Failure Archive maintenance, and emergent venture support. Speculation thus becomes a funding mechanism rather than a pure extraction. The speculative upside explicitly tied to IP Bond performance transforms passive investors into stakeholders with direct interest in Cambridge's success. A trader betting on CAM-G appreciation benefits when Cambridge ventures commercialise successfully, when IP licensing generates returns, and when spin-outs achieve exits. This alignment converts financial speculation into ecosystem advocacy.

**Service Pricing and Market Discovery** Within the Cambridge ecosystem, business support services are priced and transacted in CAM-G. The Fractional C-Suite Marketplace, Human Capital Credits system, and Connection Credits mechanism all denominate in CAM-G, creating genuine utility demand for the token beyond speculative holding. Service providers set prices through the peer-to-peer bidding mechanism, with market discovery determining equilibrium rates. Unlike a fixed-price system, this allows service costs to reflect actual supply and demand within Cambridge, with high-demand expertise commanding premium rates while abundant services remain accessible. As the ecosystem matures and support infrastructure

becomes abundant, more fractional executives become available, more laboratory capacity goes online, and more training programmes become operational, service prices will naturally decline relative to value received. Ventures find that the effective cost of accessing Cambridge’s support infrastructure becomes immaterial relative to the capability it provides, achieving accessibility through abundance rather than through artificial price controls.

#### 5.4.2 Diaspora Incentives

**Projection Over Extraction** Emotional and financial ties to origin communities generate more stable, committed capital than purely transactional foreign investment. Companies that expand internationally while maintaining active Cambridge support—for example, board seats held by Cambridge participants, create ongoing mentorship commitments to Cambridge ventures and foster research partnerships with Cambridge institutions.

Investments in subsequent Cambridge spinouts retain their CAM-G allocations as part of the Cantab diaspora, with opportunities to earn additional rewards by giving back. This establishes a Cambridge alumni network with financial and social incentives to remain engaged: departed ventures benefit from Cambridge’s continued success through token holdings, while their ongoing contributions, in turn, strengthen Cambridge’s innovation ecosystem.

**Integration with Existing Infrastructure** The token mechanism integrates with, rather than replaces, the financial infrastructure described in this document. Each component plays a distinct role:

- The **Cambridge IP Bond** provides the value anchor and appreciation driver.
- The **Cambridge Community Fund** serves as the issuance gateway and underwrites accessibility.
- The **PISCES framework** forms the foundation for the Cambridge Exchange, enabling the rest of the ecosystem to function.
- **Venture Cooperative structures** ensure collective governance over parameter decisions.
- **Evergreen Capital structures** provide patient capital, able to hold token positions across multiple venture cycles.
- The fractional **C-suite marketplace**, human capital credits, and connection credits generate service-layer utility demand.

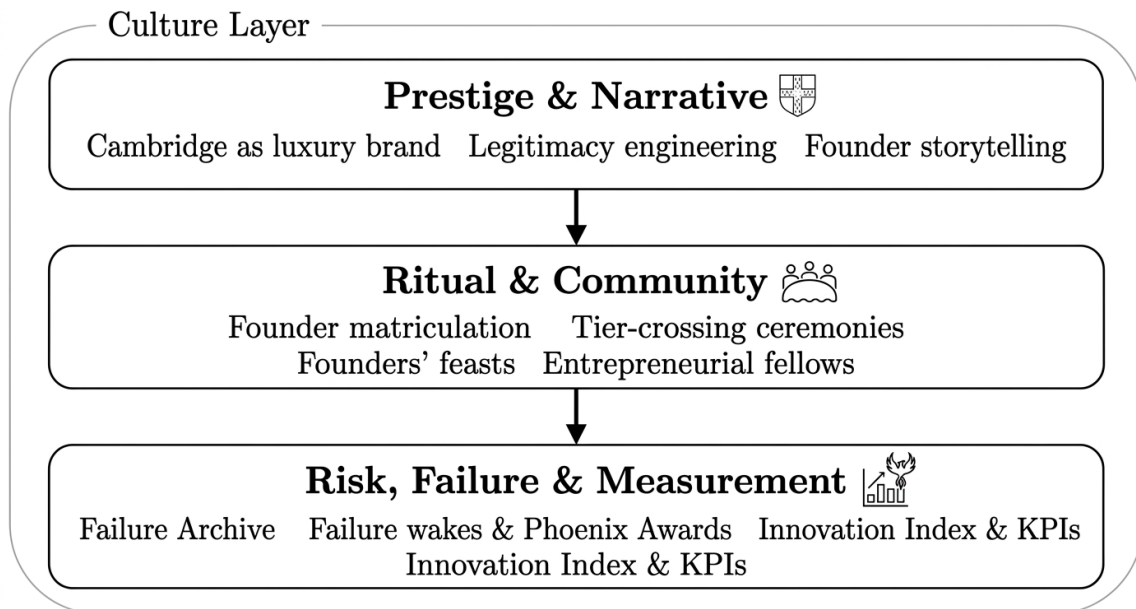
Together, these elements form a coherent financial ecosystem: the token functions as the medium of exchange, the IP Bond as the store of value, and the Community Fund as the issuance and accessibility mechanism, fulfilling the traditional functions of money within a bounded, incentive-aligned innovation ecosystem.

Overall, the Production Layer is about moving from ad-hoc venture creation to an engineered

process. By offering end-to-end support as well as a physical and a virtual home for startups, it will vastly increase the odds that Cambridge’s raw innovations become viable companies and stay in Cambridge through their growth. This layer addresses fragmentation (by centralising resources), speeds up commercialisation (with expert guidance and shared tools), and improves quality (through structured vetting and development). In manufacturing terms, we are building the “*factory for startups*”, an approach uniquely feasible in Cambridge due to its concentration of talent and resources. If successful, it could serve as a pilot model for other regions.

Much of the work of the Global Cambridge Initiative already engages with these questions. Once the criteria are satisfied, founders within the Global Cambridge programme would be granted access to investors at the point where they truly need them, rather than prematurely.

## 5.5 CULTURE LAYER



**Figure 8:** Culture Layer Interactivity

This layer treats culture as a primary governance instrument. Organisational culture is widely understood as a mechanism of social control because it shapes “*how we do things*” without requiring constant supervision or contractual enforcement (58; 65). In Cambridge, the binding forces that coordinate behaviour – prestige, tradition, status signals, and shared identity, already function as a de facto control system.

We propose a Culture Layer that is employed to grow the innovation ecosystem of Cambridge by ensuring continuity, knowledge transfer, and attractiveness of participation. At a systems level, the Culture Layer aims to operationalise Cambridge rituals, prestige mechanisms, and narrative power so that:

- entrepreneurship becomes socially legitimate and high-status (reducing psychological opportunity cost)

- ecosystem participation becomes sticky across population churn (students, postdocs, visiting fellows)
- compliance with “*hard*” mechanisms (fund contribution, data reporting, token circulation) becomes cheaper to secure because it is culturally reinforced
- innovation is integrated without brand dilution, securing the consent of incumbent institutions

### 5.5.1 Cambridge as a Luxury Brand

Culture is expensive to build, slow to change, and high-stakes to mis-handle. Our goal is to make innovation be self-generating, and for this to become an integrated element of existing institutions. Thus, the proposed innovation ecosystem should, as much as possible, correspond culturally to the existing one. In addition, we should use whatever self-driving tools that are available. Cambridge already possesses an unusually dense cultural stock (rituals, heritage, symbolic capital). The Culture Layer’s advantage is not only how it helps innovation become a core pillar of the Cambridge identity, with consent of incumbent institutions, but it reduces the time, friction, and ongoing spend otherwise required to recruit and continuously police scarce “*innovation leaders*” and contributors. Instead of relying on bespoke persuasion and heavy marketing, the system uses Cambridge’s existing legitimacy and ritual to make participation in the Training, Capital, Production, and Token/Currency layers default, prestigious, and identity consistent.

Cambridge functions in many respects as a luxury / heritage brand: it accrues value through a combination of narratives, rituals, imagery, dress, prestige, and authenticity. Luxury-brand literature is explicit that maintaining scarcity and symbolic integrity is central to preserving value, and that growth must be managed without exposing the brand to uncontrolled forces (80).

This exposes one of the key challenges of operationalising a community driven innovation culture – this imparts external entities with institutional level influence on the Cambridge brand, and dilutes the control from incumbents, such as ARM, AstraZeneca, and most importantly the University of Cambridge. Unless a case is made that it is possible to introduce innovation autonomy without brand dilution, change cannot occur.

Matching the Cambridge culture and constructing a collection of self-contained proposals that could be operated independently is the path to achieving this. The practical objective is legitimacy engineering: ensuring venture creation becomes cognitively understandable, morally acceptable, and pragmatically valuable to Cambridge’s incumbent stewards.

Legitimacy theory is useful here: for entrepreneurship to be accepted, it must become cognitively comprehensible, morally approved, and linked to audience interests (41). Since Cambridge is bottom-up, coordination must be free moving. Cultural design is the easiest way to achieve this and extends control beyond administration. The Culture Layer introduces

recurring public formats that increase shared awareness, reduce duplication, and create accountability without imposing a single authority. Thus, to achieve brand-consistent entrepreneurship, we must:

- Encode entrepreneurship as a Cambridge-grade pursuit (rigour, peer review, depth), not a deviation into “*mere commerce*”.
- Treat cultural elements (college formals, ceremonies, place, identity) as the wrapper that allows innovation to pass governance filters.

This aligns with the legitimacy framing (42): it generates pragmatic visibility, moral seriousness, and cognitive comprehensibility.

### 5.5.2 Operationalised Ritual Infrastructure

Early-stage venture-building is an uncertainty market; under uncertainty, status acts as a powerful signal that substitutes for missing information. The Culture Layer therefore formalises status pathways so that contributing to the ecosystem is not merely altruistic—it is instrumentally rational for ambitious people who care about prestige, signalling, and long-term career positioning.

Cambridge already has what most ecosystems pay to fabricate: ritualised belonging. Ritual theory explains how repeated interaction rituals generate solidarity, shared symbols, and commitment through emotional entrainment and collective attention.

The Culture Layer uses this to solve a practical scaling problem: rather than constantly searching for “*innovation leaders*” to manufacture cohesion, it must build repeatable cohort integration that generates commitment automatically, in line with existing preferences, and cheaply (relative to external recruitment).

One key attraction to participating in traditional academic culture are the ritual rites. If innovation is to be considered a viable alternative for the same pool of people, it must capture that element. Beyond simply awarding innovators with practical rewards for good performance, we must introduce symbolic rewards for progression, participation, and membership within the ecosystem. This includes induction, graduation, and regular ceremonies. We may draw from examples within Cambridge and academic communities at large for inspiration. A matriculation ceremony can accompany the signing on of a company to a given portfolio, with a physical book and archivist to preside over this. A graduation ceremony, with processions, awards of certificates, and dress, can accompany the completion of incubators, accelerators, or even the closing of a funding round. Regular formal dinners of colleges, each with quirks and traditions and customs, can be part of the life afforded members of the Cambridge innovation community.

Following on from the collegiate fellowship model, entrepreneurial fellows can be employed in fixed-term contracts, focused on mentoring potential innovators, constructing datasets, reinvesting in Cambridge ecosystem innovation, etc. In this, they obtain collegiate rights,

including status, affiliation, and dining or event benefits, all of which convert to high quality networking and signalling. These can be through negotiating partnerships with colleges, expanding out existing innovation or entrepreneurial fellowships, or it can be independent.

This can be integrated with founder/operator in residence roles from the production layer as a pool from which to sample entrepreneurial founders, acting as an incentive to participate. Entrepreneurial fellows may further function as managing founders/operators in residence, listed as a responsibility and benefit of the role.

- **Entrepreneurial Fellows:** time-bounded affiliations attached to colleges/departments, awarded for measurable contribution (mentoring hours, data contribution, reinvestment).
- **Founder-in-Residence / Operator-in-Residence rotations:** institutional roles positioned parallel to visiting fellows, embedding commercial capability inside Cambridge legitimacy structures.
- **Tutor-level signalling:** equip academic gatekeepers to treat entrepreneurship as “*Cambridge-valid*”, reducing identity-risk at the point of decision.

This mechanism is explicitly designed to replace brute-force recruitment: instead of hunting for leaders, the system creates a ladder where high-quality contributors self-select because the status incentives are aligned with Cambridge’s prestige economy.

Each of these may also leverage existing facilities. Cambridge has numerous buildings and spaces of exceptional beauty and history, within which events can be run. College partnerships can include opportunities to dine within college, or a calendar of formal dinner reservations can cycle through colleges. At a smaller scale, regular meetups of innovation players in Cambridge (unnamed to respect exclusivity) locations already exist, and the regularity and community generated by these can be the basis for more high-end events. Some example events thus include:

- **Founder Matriculation:** entry ceremony into the Venture Foundry pipeline (rotating colleges; cohort identity).
- **Tier Crossing Ceremonies:** public rites of passage when ventures pass certain phases in their growth (ideation → incubation → acceleration), mirroring academic progression.
- **Founders’ Feast / College Venture Formals:** termly ritual gatherings with designed formats (rotating prompts, “*problem auctions*”), turning social tradition into production-relevant collisions.

While events, symbols, and signalling can serve as forms of ritualisation of innovation, group events can be formalised to optimise for functional outcomes. We propose this take the form of a 3-function model that formalises “*designed collisions*”, with events each designed to preserve Cambridge’s decentralised character. Institutional theory explains that stable fields reproduce themselves through normative and cognitive pressures; if cross-silo exchange is not institutionalised, fragmentation persists by default.

This is broken down into three functions – structured, unstructured, and driven networking. Structured networking formally introduces people to one another to expedite initial connection with as many as possible, unstructured networking allows them to interact freely and establish independent bonds, and driven networking connects people on tasks, generating resilient, work-based bonds. These feed forward into community and innovation generation for the Production Layer.

Events with structured mixing and with regular, dynamic, free, unstructured events are necessary for the innovator population to produce personal networks and become entrenched in the ecosystem. This allows them to navigate the ecosystem, makes them invested within it, and permits them to derive value from it even during low capital return periods. Connecting them along the same channels with work applies this to the group level, not just the individual one. This enables resilience to business cycles or disruptions, allowing commitment to an ecosystem through these periods. Such matters are commonly available in Cambridge, though there is scarce central communication and data sharing for all three to make them grow one another. In addition, driven networking as a common thing is practically absent in Cambridge, instead relying on mixers from sandboxes or particular incubators, with limited use as they tackle the core problems of that sandbox or incubator.

Structured networking can include role-based mixers keyed on particular functions (regulation, product, hiring, etc.), speed networking events that cycle juniors through senior innovators, or failure roundtables where people discuss one business failure they made and break down barriers between one another. Further events can be designed and proposed as organisers have different interests or capabilities.

For unstructured networking, there are many options. Leveraging knowledge capital in Cambridge, high quality lectures are easy to organise, anchoring a common topic of discussion. Facility visits to allow innovators to find useful spaces for themselves. making use of plentiful third spaces in Cambridge while driving usage. Leveraging Cambridge's beauty, regular walking tours or tourist site visits can attract attendees for entertainment value, and bond over shared memories. These can be combined for walking lectures to bond people on common topics during the entertainment experience, though this is a more complex experience to do well. Pub or similar space visits are standard parts of such events or even standalone events.

For driven networking, these can be quite imaginative. Problem salons anchored on particular issues experienced within the network and to extract community insight into solving them are one option. Hackathons are classic examples. Group clinics from established business service providers to generate client/provider and co-attendee relationships while practically serving businesses. In general, any event should have a guiding task and force attendees to witness and be direct beneficiaries of fellow attendees' capabilities. This establishes trust and problem bonding.

- 3 phases of connection building: unstructured, structured, and driven networking.

- Salons and “*problem dinners*”: curated, mixed tables (lab founders, MBA talent, angels, domain academics, operators) with explicit outputs (introductions, pilots, evaluations).
- Cambridge Question Time: recurring forum where founders, academics, investors, and civic leaders address barriers and priorities.
- Facility crawls / open-house rituals: repeated tours positioned as calendar staples, making exploration and collaboration a norm rather than an exception.
- Role-based mixing: function-specific events (regulatory, product, hiring, sales) designed to feed directly into Production Layer needs.
- Group experience events: walking tours as structured group bonding (free, high signal, Cambridge-native), used as a standard onboarding instrument.

### 5.5.3 Combining Risk Taking with Academic Culture

The Culture Layer involves rituals, status, and culture which matches existing Cambridge academia, applied to the innovation process. This gives signalling type rewards for participation as a valuable growth and culture sustaining mechanism, but innovation requires failure, and “*fail fast*” culture collapses when reputational penalties are high. Innovation requires failure, but this proposed status economy makes failure reputationally costly. Therefore, we run the danger of collapsing innovation learning unless we build protected disclosure rituals and system memory. Research on psychological safety shows that learning behaviour depends on environments where interpersonal risk-taking is safe. Complementarily, “*intelligent failure*” work argues that certain classes of failure are productive and should be systematically cultivated.

Active efforts should be implemented to make failure an accepted concept, part of the innovation process, and the mechanism by which learning occurs. This is such that innovation failures are not considered social failures and innovators are comfortable putting themselves at real risk since they recognise the learning arbitrage of the attempt. One classic challenge of achieving this is self-confidence that learning is not necessary for an innovator. Self-confidence is an important driver of someone towards innovation, and must not be broken – instead, effort should be made to clarify that the correct stance, the one which leads to success, is the ongoing necessity of improvement, and thus education. Normalisation of failure from peers and seniors should allow for self-confidence to become resilient to it, allowing innovators to more effectively practice lean startup creation, where attempted implementation and its failure is the key mechanism by which strategy is developed, ensuring maximal exposure to market and thus maximising likelihood of product-market fit.

Options include datasets, such as a failure archive, an anonymised dataset of local company failures and data concerning it, fed by “*failure wakes*”, confidential, facilitated forums paired with structured post-mortems from which data is catalogued.

Infrequent, regular events featuring high-quality learning contributions and intelligent failures can work by giving real rewards of tangible value to winners, making disclosure respectable rather than shameful along with the failure wakes. These can be branded as the “*Phoenix Awards*” as part of a push to rebrand failure as knowledge capture via the entrepreneurial process.

- **Failure Wake:** confidential, facilitated forum paired with structured post-mortems feeding a Failure Archive (privacy-preserving).
- **Phoenix Awards:** annual recognition for high-quality learning contributions and intelligent failures, making disclosure high-status rather than shameful.

#### 5.5.4 Integrating Culture with other Layers

As a means to ensure we have insight into how governance actually operates, recall that culture stabilises systems by making behaviour legible and shared. The Culture Layer therefore produces a small number of “*public artefacts*” that function as coordination objects—what people point to, debate, and align around.

Events and communications should be designed in ways, each time, to ensure a core narrative is promoted – that one should learn from one another, that we should form teams quickly and openly, that we should build solutions for the rest of the community, that they should leverage the community to fund these solutions, and that they should reinvest within it. In general, each communication should be checked to promote at least one of these ideas, not contradicting any. If contradictions occur, then work should be made to reframe them within the narrative. For example, if suggestions are made to seek outside funding, then messaging should promote incorporating outside entities into the innovation community.

Any indices on the community should be taken care to be promoted, discussed, and advertised as indicators of community quality. They should make members proud to have contributed to such measures, and prompt investigation into the cause for such good performance. Negative performance should be framed as coming from outside conditions, paired with steps to mitigating these, and ideally reframed as opportunity.

Finally, ventures, initiatives, and services by members of the innovation community should be promoted intensely. Formal channels with community symbols should be established to allow members to advertise, affect opinion, and establish themselves as thought leaders. In particular, excellence and rigour should be key aspects projected in all communications and conduct, following the brand construction of the University of Cambridge.

Unifying these, ongoing brand and communication monitoring should be established, with clear brand guidelines for different functions, working at both strategic (for scheduling and redirection) and operational (for brand dos and don’ts) levels. Following the language of optimisation, the objective to optimise for is community championing, while other metrics should be treated as constraints, to achieve sufficiency to a certain degree. This could be as

simple as, “*at least a 4 out of 5*” as personal culture manager opinion.

- A single common narrative linking the five layers: “*learn → form teams → build → fund → reinvest*”.
- Monthly Cambridge Innovation Index: positioned as a cultural artefact (discussed, referenced, identity-forming), not merely a report.
- Founder-facing storytelling: formalised amplification of ventures as Cambridge-normal trajectories.
- Public dashboards (selected metrics): make ecosystem health culturally discussable and legible.

Cultural framing determines whether financial mechanisms feel like stewardship or extraction. The Culture Layer therefore supplies the interpretive layer that makes contributions, data sharing, and token/currency circulation acceptable inside Cambridge’s prestige code.

All efforts must be framed as Cambridge-continuous, an extension of scholarship, rigour, and public value—rather than as a foreign accelerator aesthetic. It must be communicated that community members are operating to a structured brand policy, that it is practically advantageous for them to do so (and thus are likely to follow it independently), and that this has been clearly signalled to and understood by them.

Functional means of implementing this at scale for several independent initiatives would be the introduction of cultural consultancy services, who, following a defined brand book, can focus on ensuring commonality among those companies who use culture to extend their assets. The advantage of this is that this will permit those companies to remain independent as they wish, while projecting themselves as a singular entity, attracting top talent and IP in the process, thus collecting them together in the same broader group.

- Frame community contributions as membership obligations (stewardship), not taxation.
- Frame token/currency circulation as infrastructure coordination, not gimmickry.
- Couple economic rewards with status recognition (titles, ceremonies, public acknowledgement), so compliance is reinforced socially as well as financially.
- Define a singular cultural brand book, accessible to all members of the Cambridge innovation community.

The outcome of this is that the Training Layer becomes socially endorsed, the Capital Layer becomes morally normalised, and the Production Layer becomes a recognised civic institution. This is all without singular leadership, rather the Culture Layer reduces dependence on scarce individuals by making the system robust. Instead of repeatedly paying to search for and incentivise exceptional people, the Cultural Layer invests in the Cambridge Innovation ecosystem such that it:

- attracts high-quality participants through prestige and identity alignment (status signalling under uncertainty)
- reduces enforcement and coordination cost by substituting norms and legitimacy for continual monitoring
- creates continuity across churn because rituals outlive champions (culture as social control)
- increases cross-silo collaboration density by institutionalising exchange (normative/cognitive field effects)
- improves learning rate and founder retention by making intelligent failure safe and productive

In summary: the Culture Layer is the implicit control mechanism that allows the other layers to operate with consent, legitimacy, and scalable behavioural adoption, because it aligns system participation with the deepest currencies in Cambridge: prestige, tradition, and belonging.

## 6 CONCLUSION

Cambridge faces a fundamental paradox. It produces world-class science and has created Europe’s largest technology cluster, yet the economic gains from its outputs do not match their global significance. Our analysis highlights interconnected structural challenges, rather than isolated failures. The central question is how to upgrade the ecosystem without sacrificing what makes Cambridge unique.

Cambridge’s bottom-up ecosystem drives innovation and dynamism, but it also produces fragmentation and discontinuity. Siloing can limit collaboration, yet it reflects the city’s distinctive ideation process. Academic incentive structures promote best-in-class intellectual property but can constrain entrepreneurial growth. Many ideas that bridge research and business struggle to scale under these conditions, despite numerous initiatives attempting to support innovators. The result is a hidden opportunity cost for founding and growing ventures in Cambridge, which often discourages retention.

To mitigate this, we propose the Cambridge Startup Production Model, a coordinated, four-layer system designed to strengthen the ecosystem without disrupting its organic strengths. The **Training Layer** would establish entrepreneurship as a recognized academic pathway, introducing PhD by Innovation tracks, career routes where commercial impact contributes to promotion, and structured programs that match technical talent with business expertise. The **Capital Layer** would provide community-driven funding and evaluation mechanisms, including a Cambridge Community Fund, IP-backed bonds, a local stock exchange within the PISCES framework, and reformed investment decision processes leveraging Cambridge’s concentrated expertise. The **Production Layer** would function as a venture foundry, offering shared services, tiered progression gates, a Failure Archive to capture institutional learning, and

dynamic mapping of ecosystem resources. Finally, the **Token Layer** would align individual incentives with collective ecosystem success through a Cambridge Growth Token, ensuring circulation via asymmetric vesting and maintaining accessibility for new entrants.

Together, these layers aim to reduce fragmentation, accelerate commercialization, improve deal quality, and create a self-reinforcing cycle that retains talent, fosters innovation, and preserves the unique character of Cambridge while maximizing the economic and social value of its world-class science.

## 6.1 Recommended Next Steps

Immediate priorities should focus on validating the model's components. This means conducting detailed feasibility studies for the Cambridge IP Bond and Community Fund structures, piloting the PhD by Innovation pathway with two or three willing colleges, establishing the Cambridge Failure Archive to begin capturing institutional memory, and developing the shared data platform that would underpin investment decision reform.

Medium-term implementation would involve launching the fractional C-suite marketplace with a small cohort, formalising the Cambridge Evaluation Protocol with domain experts across key departments, creating the dynamic front-door mapping of ecosystem resources, working with the Financial Conduct Authority on Cambridge-specific applications of PISCES infrastructure, and starting development of the Cambridge Token architecture in line with the development of other initiatives.

Longer-term institution building requires establishing the Research-Commercial Synthesis entity with dual academic and commercial reporting, developing Cambridge-specific ecosystem metrics and a monthly innovation index, constructing the physical innovation hub that would anchor the Production Layer, and building the international partnerships that allow Cambridge ventures to access global markets without relocating.

## 6.2 What This Could Lead To

If implemented systematically, this model could transform Cambridge from a place that incubates innovation for others to capture into a self-sustaining engine that retains and compounds its own success. Practical outcomes would include higher founder retention as opportunity costs are reduced and local capital becomes more accessible, faster commercialization as structured pathways replace reliance on serendipity, improved deal quality as ventures receive guidance before incorporation, and stronger network topology as productive connections are deliberately cultivated. Additionally, the concentration of innovation activity within Cambridge would likely create a valuable, accessible source of information and context, enabling new research into innovation science while leveraging the city's research capital, thereby stimulating global innovation.

More fundamentally, Cambridge could become a demonstration case for how academic

innovation ecosystems can be systematized without sacrificing their organic character. The model preserves decentralization and opt-in participation while providing the connective tissue and institutional memory that bottom-up systems often lack. Success here would offer a replicable template for other research-intensive cities facing similar retention and growth challenges.

The stakes extend beyond Cambridge itself. The UK's ability to translate scientific excellence into retained economic power, and the lessons this provides for global knowledge economies, depends on whether innovation hubs like Cambridge can solve the innovation paradox. This report offers a framework for doing so and represents a step toward constructing a formal, reproducible model for innovation ecosystem development.

We finish with one open question: is Cambridge's difficulty in retaining and scaling ventures truly a "*scaling problem*," or does it reflect the UK's broader innovation model, in which ideas are born locally but scale naturally beyond national borders? Is this pattern a structural weakness to be addressed, or a strategic design in which the UK specializes in ideation, research, and early venture formation, while later-stage growth occurs elsewhere? We intentionally leave this unresolved, inviting readers to reflect on their experiences and assumptions, as their insights, critiques, and interpretations will be essential as this work evolves.

For any feedback, reflections, or contributions, please contact: [policy@globalcambridge.org](mailto:policy@globalcambridge.org)

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